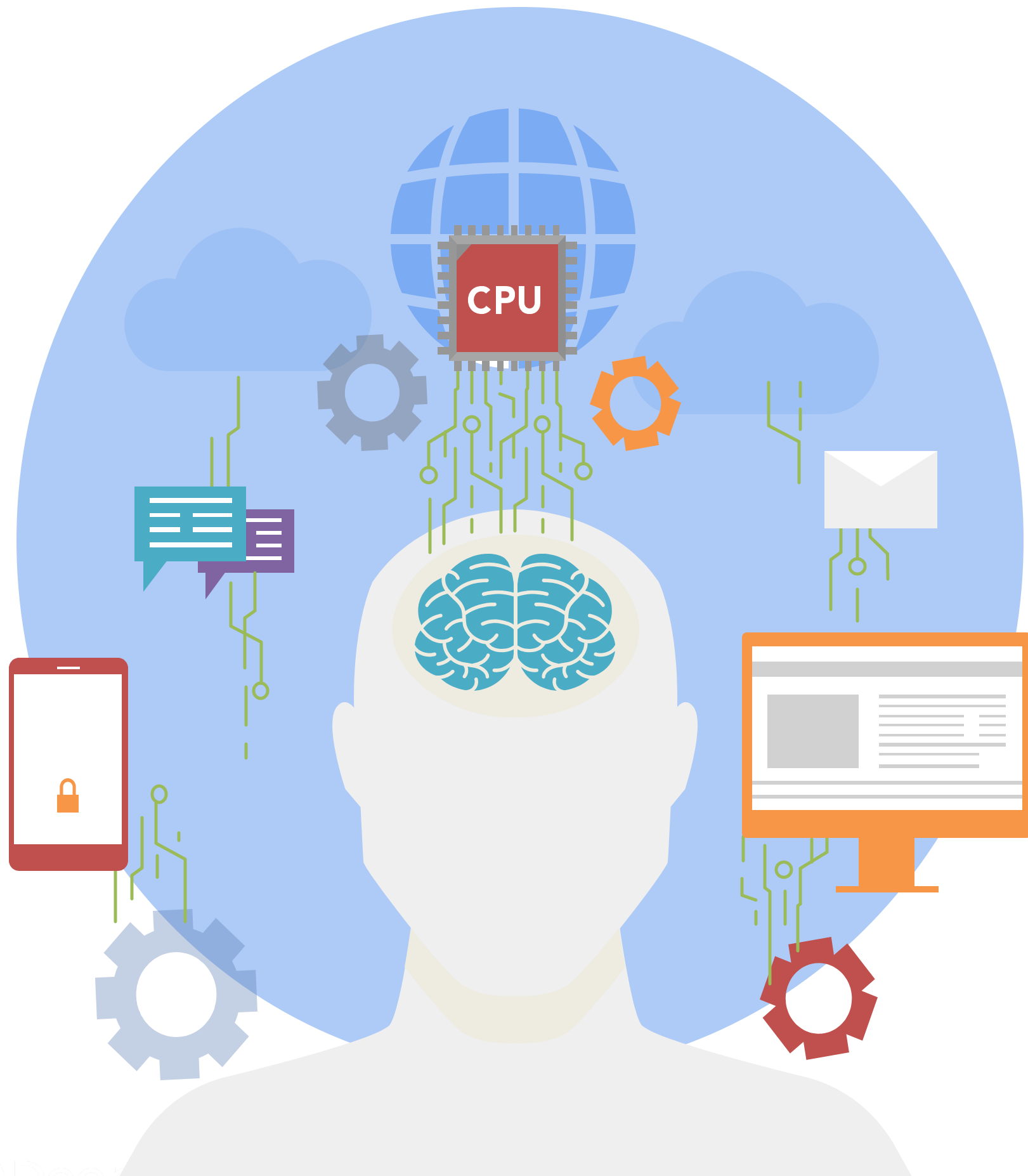




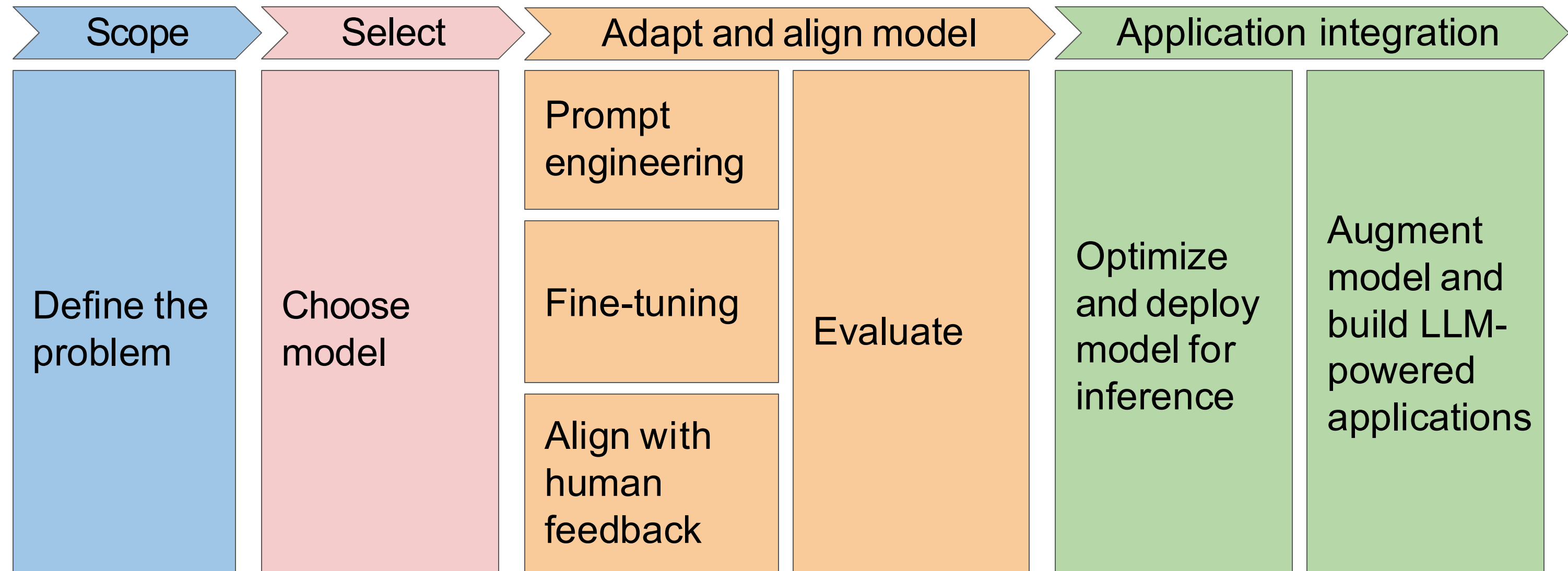
Fine-tuning, Instruction Tuning and PERF

Dr. Tamer Arafa

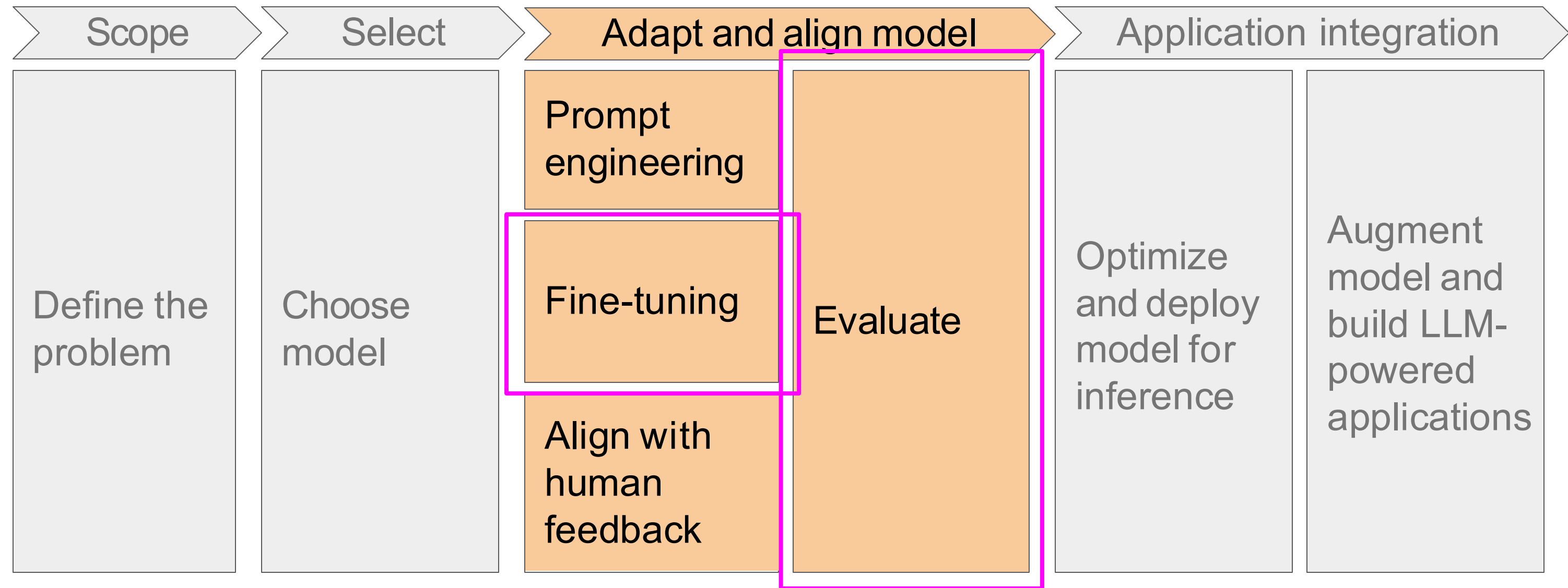
Fine-tuning with instruction prompts



GenAI project lifecycle

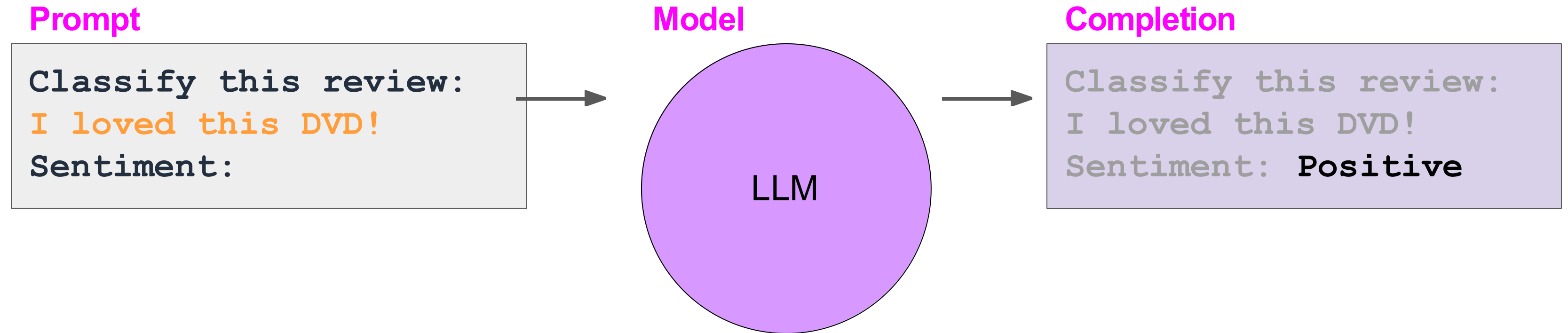


GenAI project lifecycle

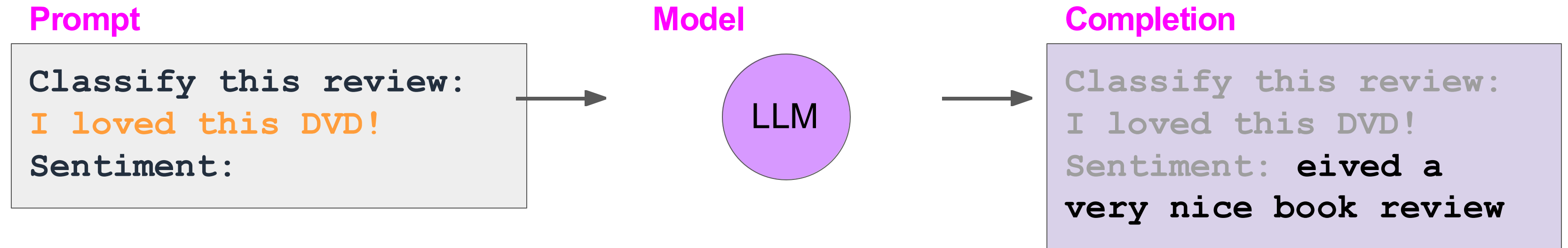


Fine-tuning an LLM with instruction prompts

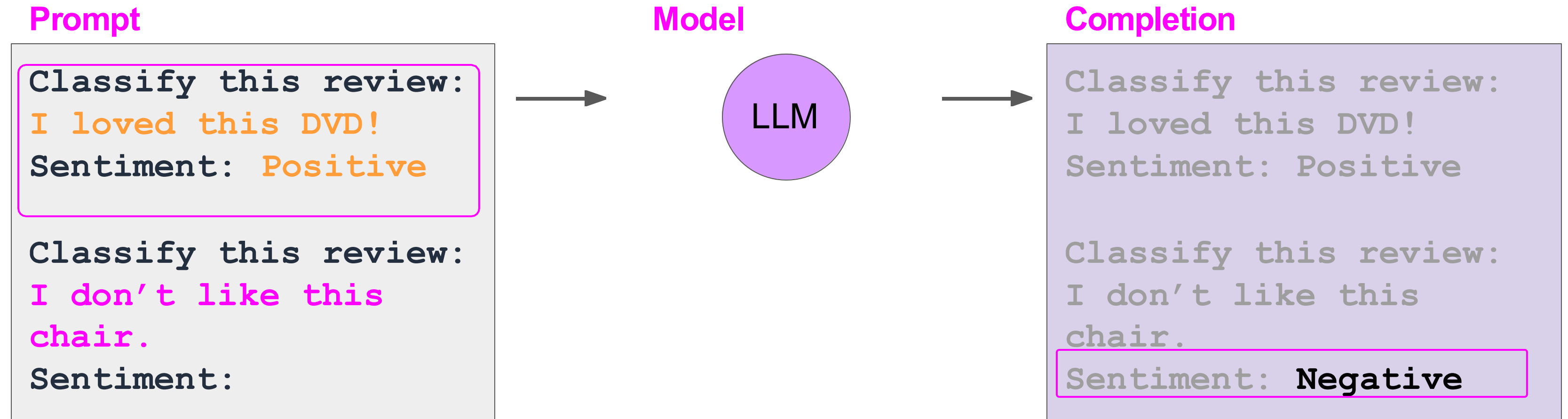
In-context learning (ICL) - zero shot inference



In-context learning (ICL) - zero shot inference

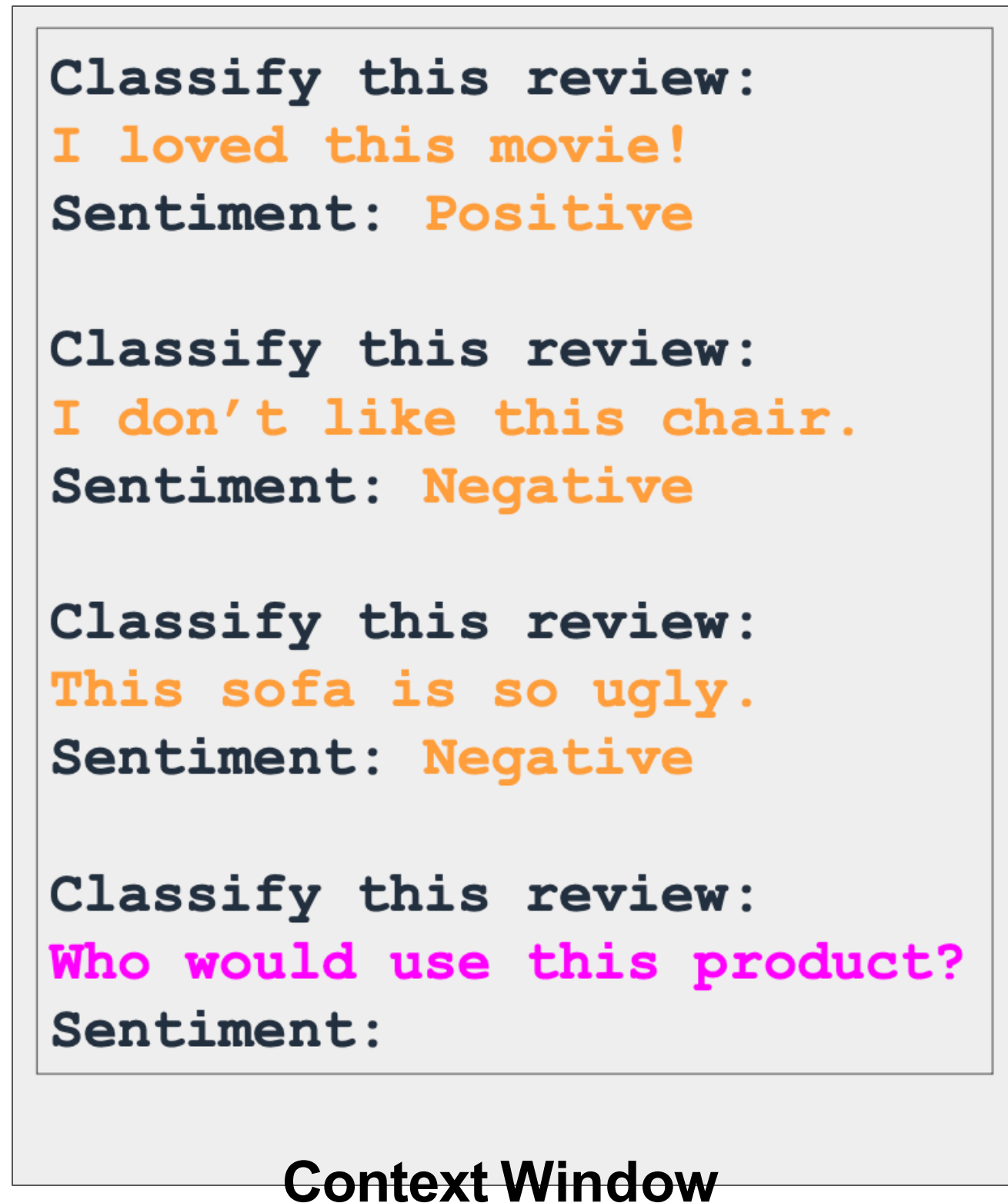


In-context learning (ICL) - one/few shot inference



One-shot or Few-shot Inference

Limitations of in-context learning



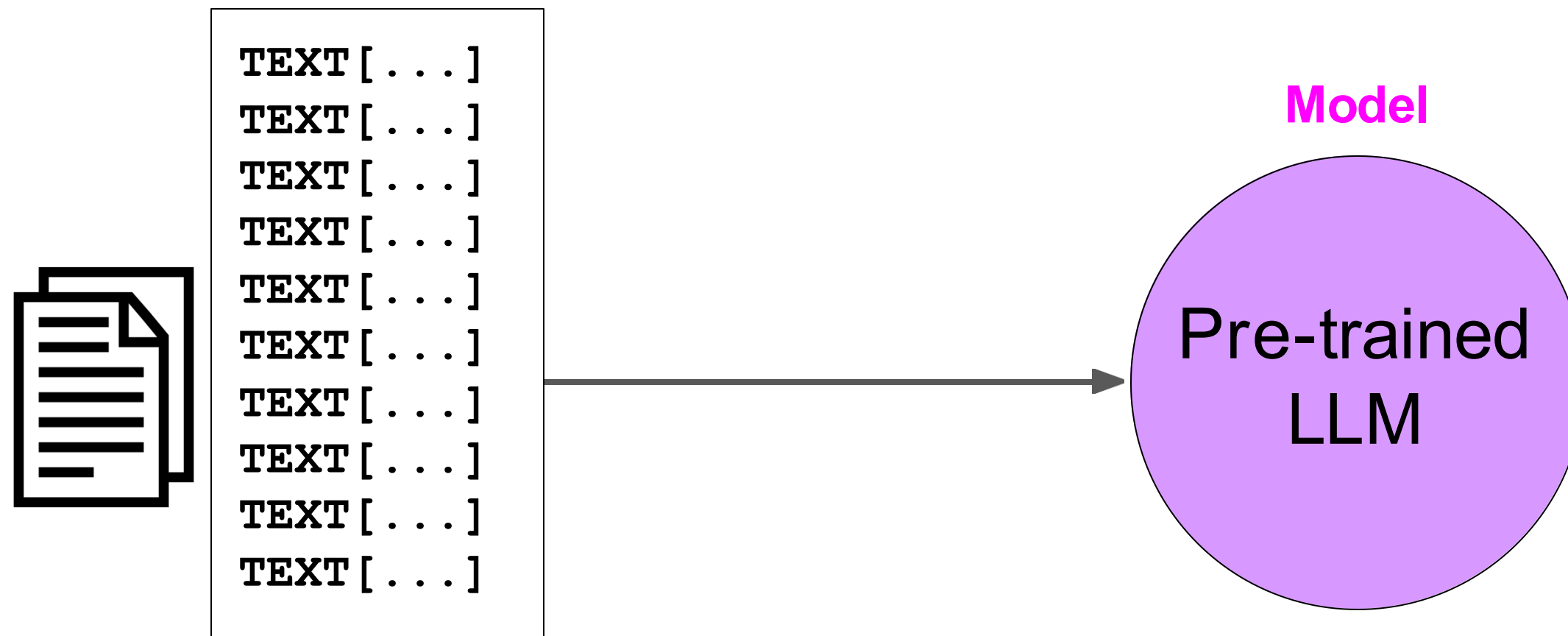
Even with
multiple
examples

- In-context learning may not work for smaller models LLM
- Examples take up space in the context window

Instead, try **fine-tuning** the model

LLM fine-tuning at a high level

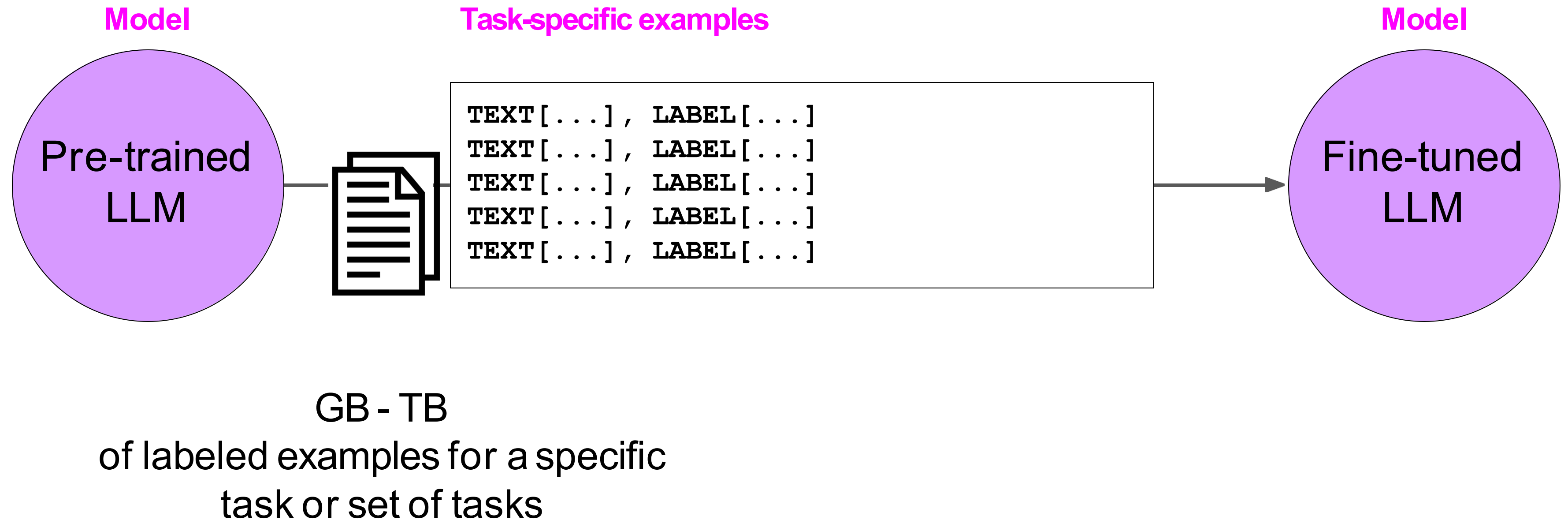
LLM pre-training



GB - TB - PB
of unstructured textual data

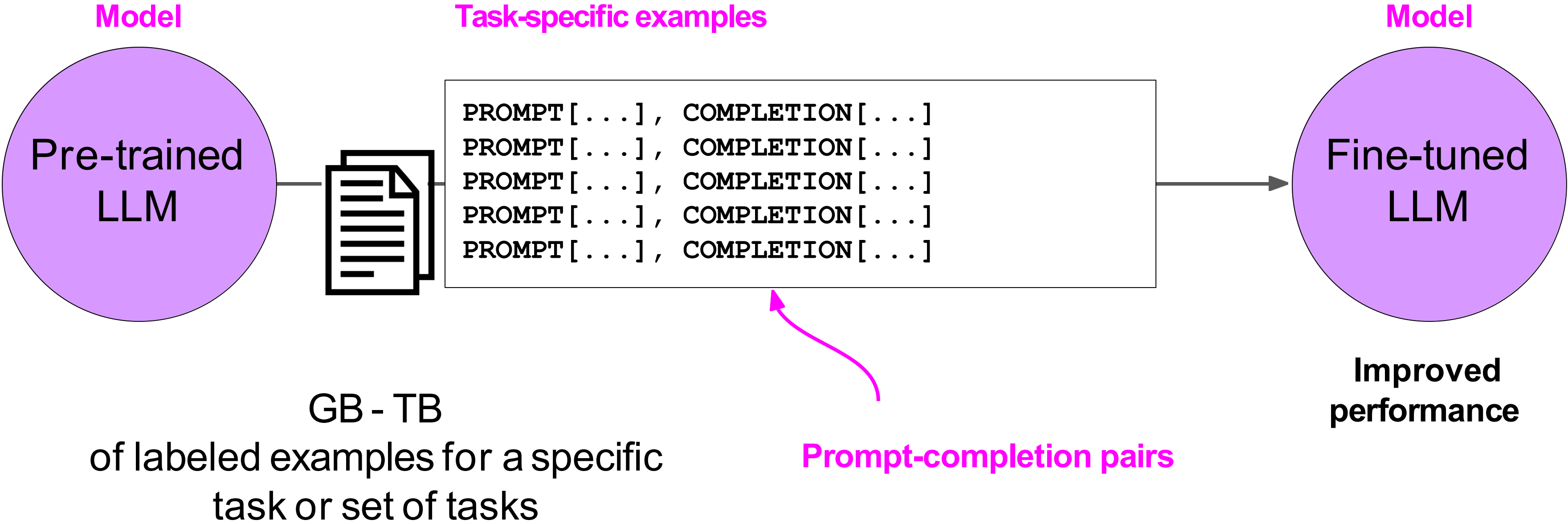
LLM fine-tuning at a high level

LLM fine-tuning



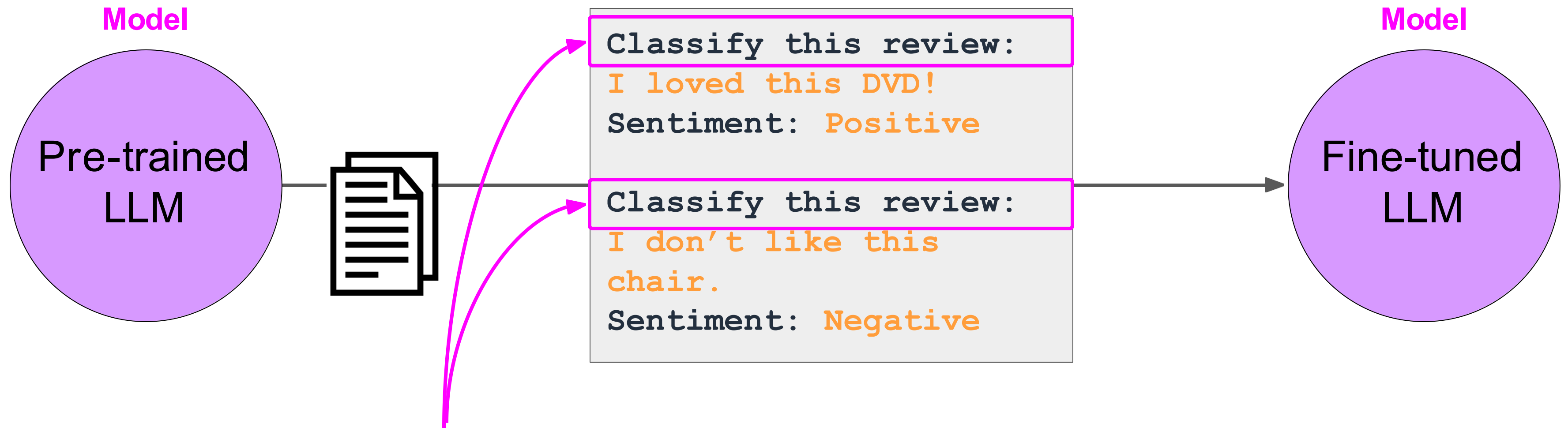
LLM fine-tuning at a high level

LLM fine-tuning



Using prompts to fine-tune LLMs with instruction

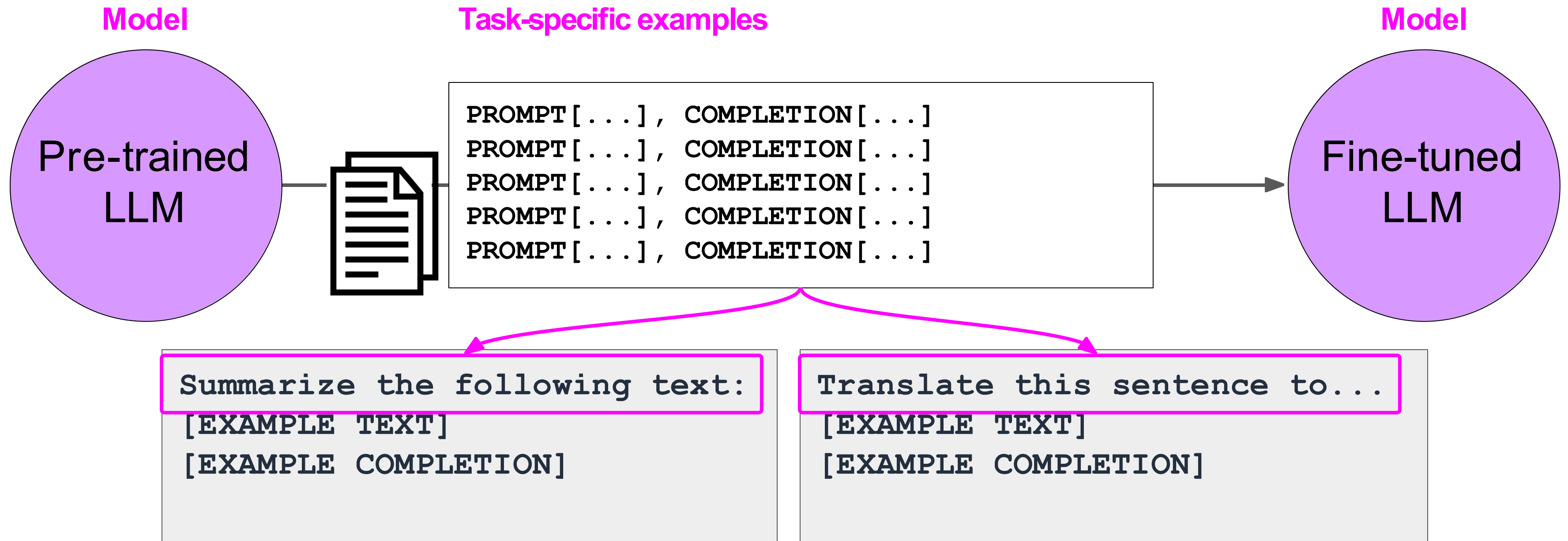
LLM fine-tuning



Each prompt/completion pair includes a specific “instruction” to the LLM

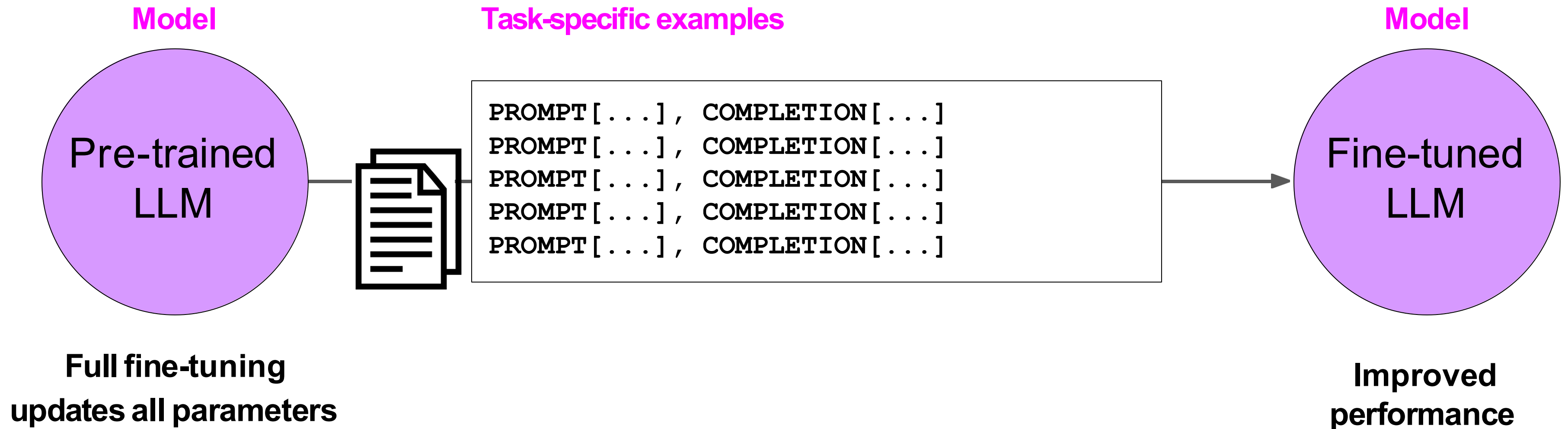
Using prompts to fine-tune LLMs with instruction

LLM fine-tuning



Using prompts to fine-tune LLMs with instruction

LLM fine-tuning



Sample prompt instruction templates

Classification / sentiment analysis

```
jinja: "Given the following review:\n{{review_body}}\n\npredict the associated rating\n\nfrom the following choices (1 being lowest and 5 being highest)\n- {{ answer_choices\n  \ | join('\n- ') }}\n\n|||\n\n{{answer_choices[star_rating-1]}}"
```

Text generation

```
jinja: Generate a {{star_rating}}-star review (1 being lowest and 5 being highest)\n\nabout this product {{product_title}}. ||| {{review_body}}
```

Text summarization

```
jinja: "Give a short sentence describing the following product review:\n{{review_body}}\n\n\n|||\n\n{{review_headline}}"
```

Source: https://github.com/bigscience-workshop/promptsources/blob/main/promptsources/templates/amazon_polarity/templates.yaml

LLM fine-tuning process

LLM fine-tuning

Prepared instruction dataset



Training splits

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]
```

Training

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
...
```

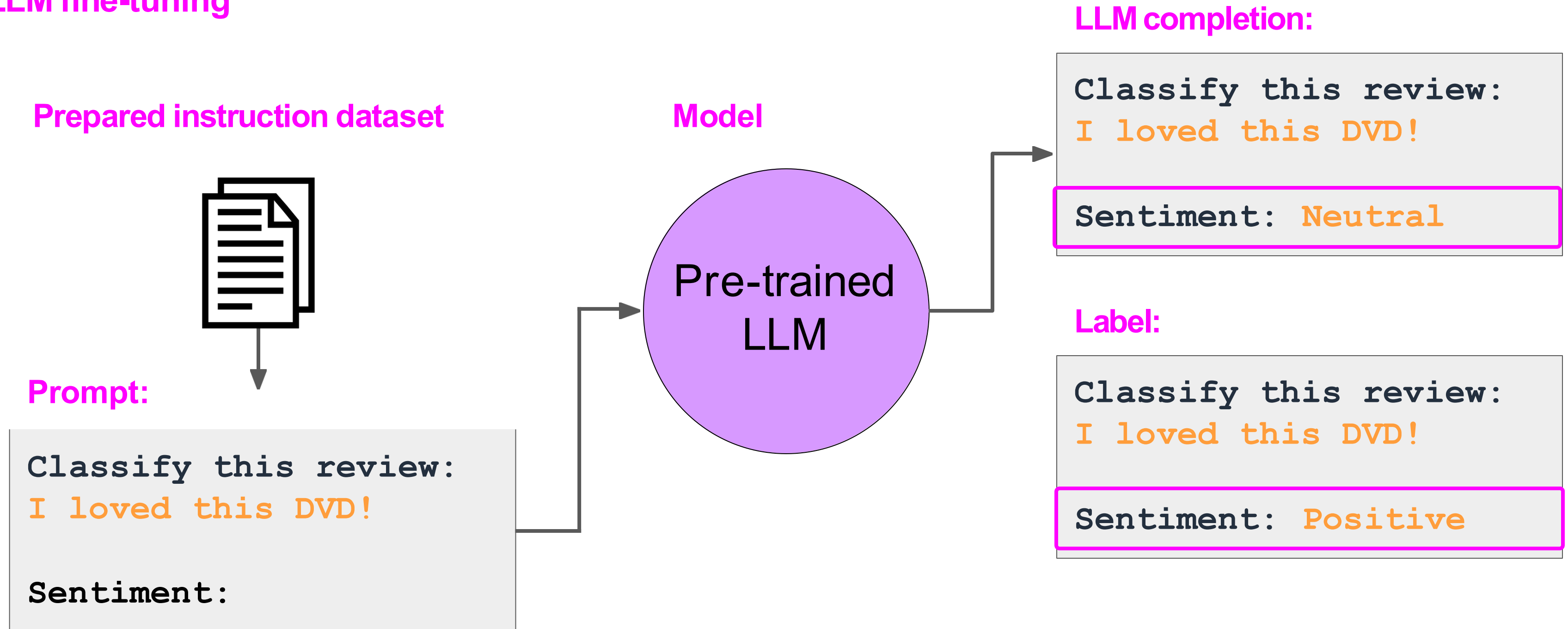
Validation

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
...
```

Test

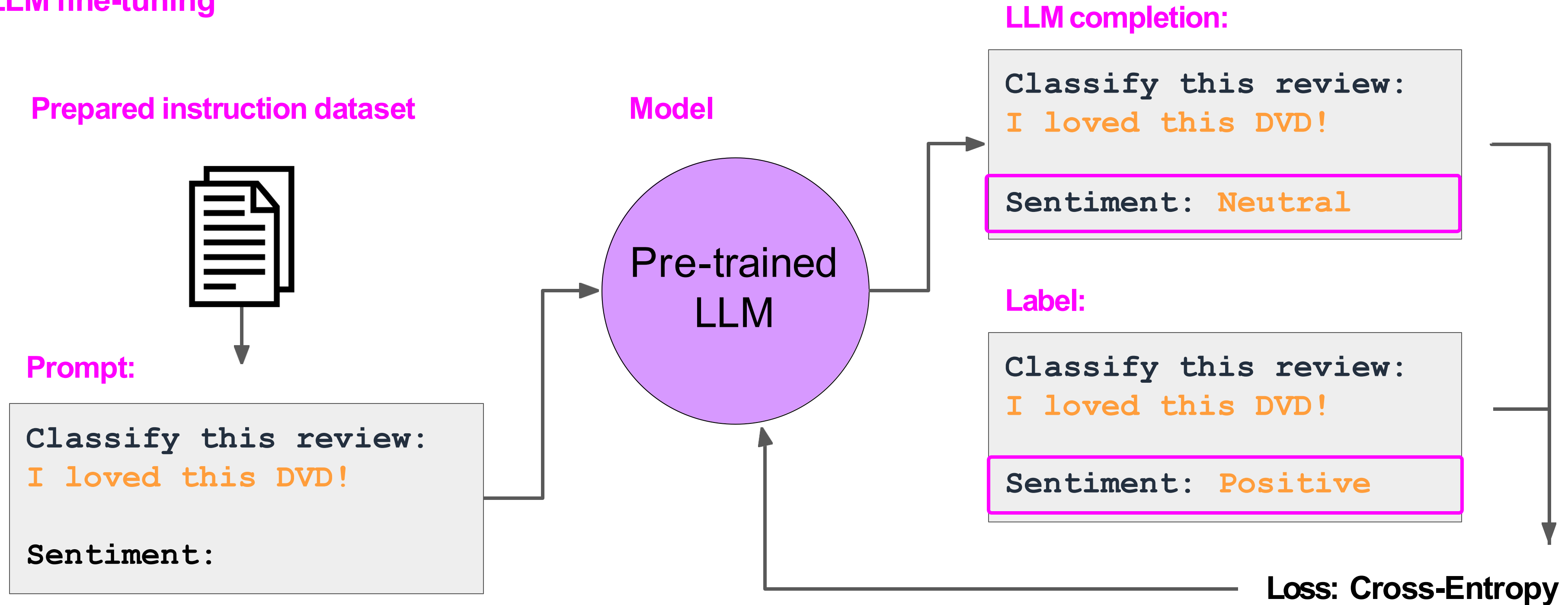
LLM fine-tuning process

LLM fine-tuning



LLM fine-tuning process

LLM fine-tuning



LLM fine-tuning process

LLM fine-tuning

Prepared instruction dataset



Training splits

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]
```

Training

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
...
```

Validation

validation_accuracy

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
...
```

Test

LLM fine-tuning process

LLM fine-tuning

Prepared instruction dataset



Training splits

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]  
PROMPT [ . . . ] , COMPLETION [ . . . ]
```

Training

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
...
```

Validation

```
PROMPT [ . . . ] , COMPLETION [ . . . ]  
...
```

Test

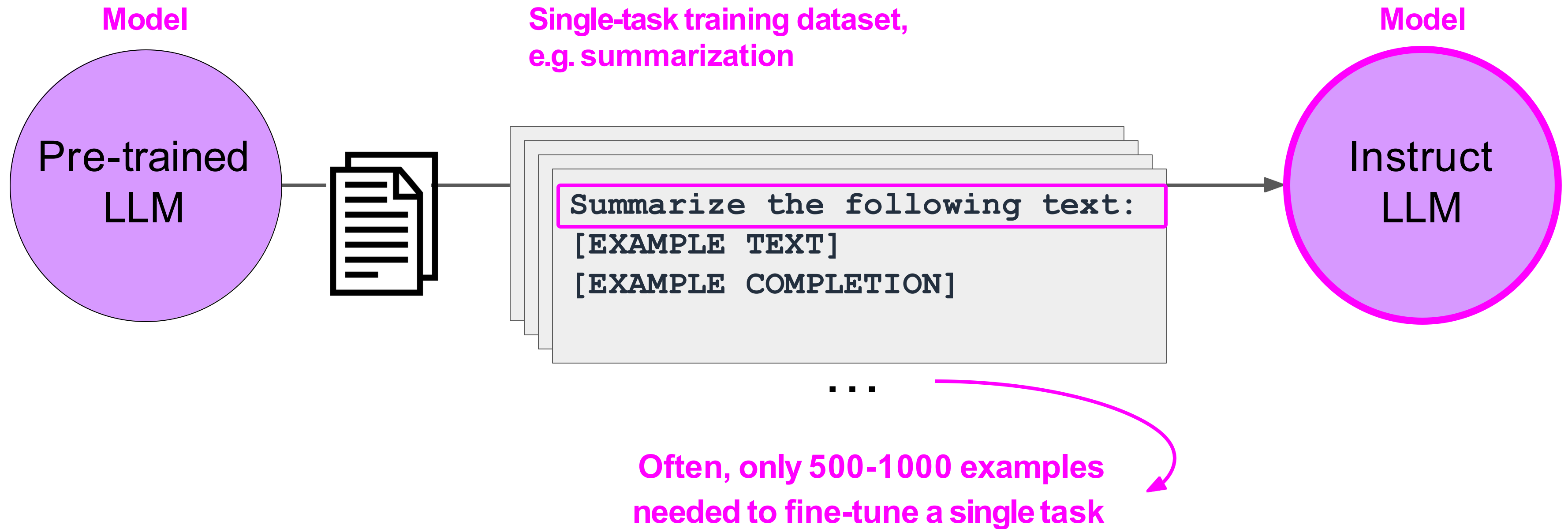
test_accuracy

LLM fine-tuning process



Fine-tuning on a single task

Fine-tuning on a single task



Catastrophic forgetting

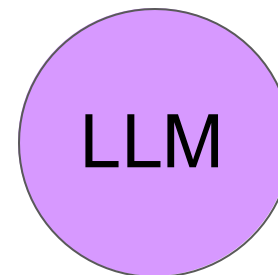
- Fine-tuning can significantly increase the performance of a model on a specific task...

Before fine-tuning

Prompt

Classify this review:
I loved this DVD!
Sentiment:

Model



Completion

Classify this review:
I loved this DVD!
Sentiment: eived a
very nice book review

Catastrophic forgetting

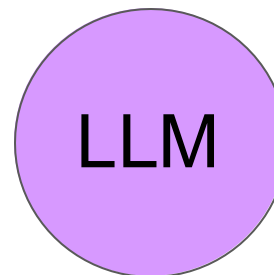
- Fine-tuning can significantly increase the performance of a model on a specific task...

After fine-tuning

Prompt

Classify this review:
I loved this DVD!
Sentiment:

Model



Completion

Classify this review:
I loved this DVD!
Sentiment: **POSITIVE**

Catastrophic forgetting

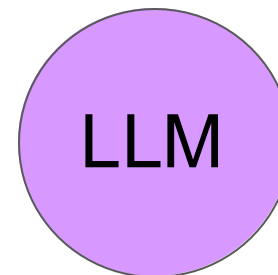
- ...but can lead to reduction in ability on other tasks

Before fine-tuning

Prompt

What is the name of
the cat?
Charlie the cat roamed
the garden at night.

Model



Completion

What is the name of
the cat?
Charlie the cat roamed
the garden at night.
Charlie

Catastrophic forgetting

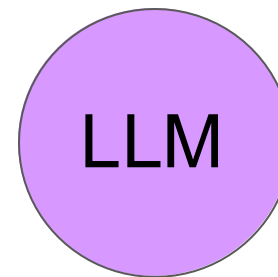
- ...but can lead to reduction in ability on other tasks

After fine-tuning

Prompt

What is the name of
the cat?
Charlie the cat roamed
the garden at night.

Model



Completion

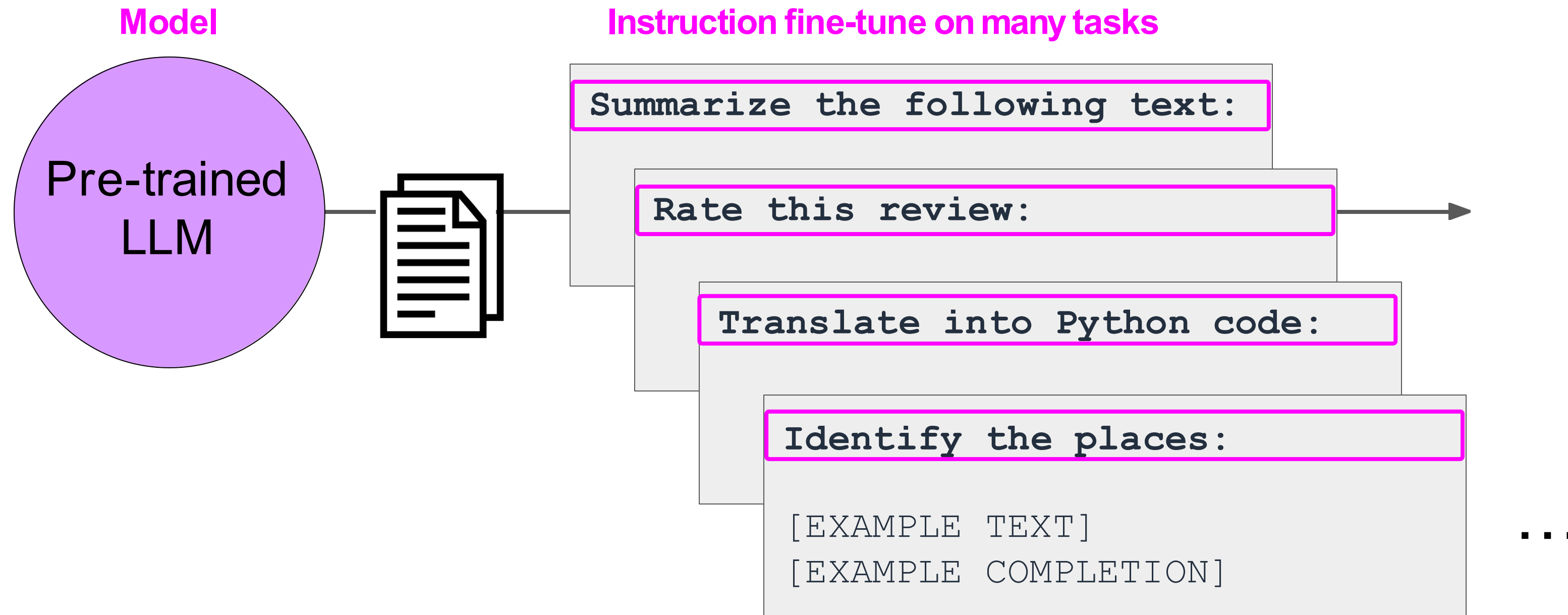
What is the name of
the cat?
Charlie the cat roamed
the garden at night.
**The garden was
positive.**

How to avoid catastrophic forgetting

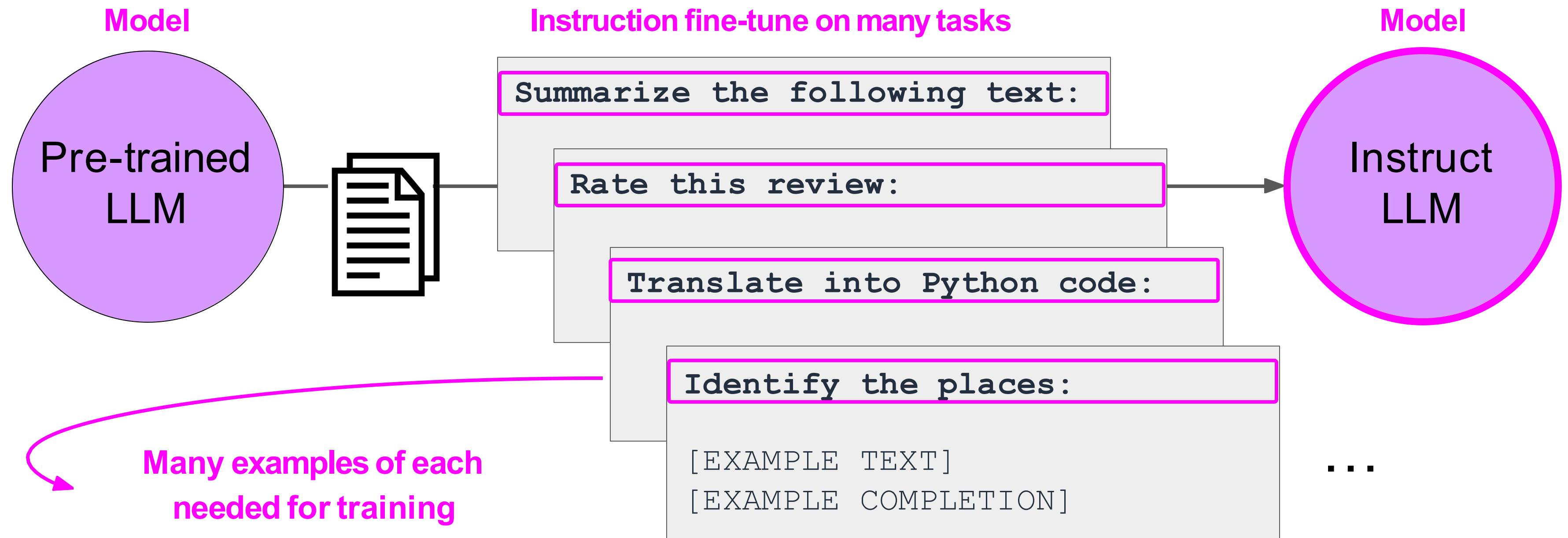
- First note that you might not have to!
- Fine-tune on **multiple tasks** at the same time
- Consider **Parameter Efficient Fine-tuning** (PEFT)

Multi-task, instruction fine-tuning

Multi-task, instruction fine-tuning



Multi-task, instruction fine-tuning



Instruction fine-tuning with FLAN

- FLAN models refer to a specific set of instructions used to perform instruction fine-tuning

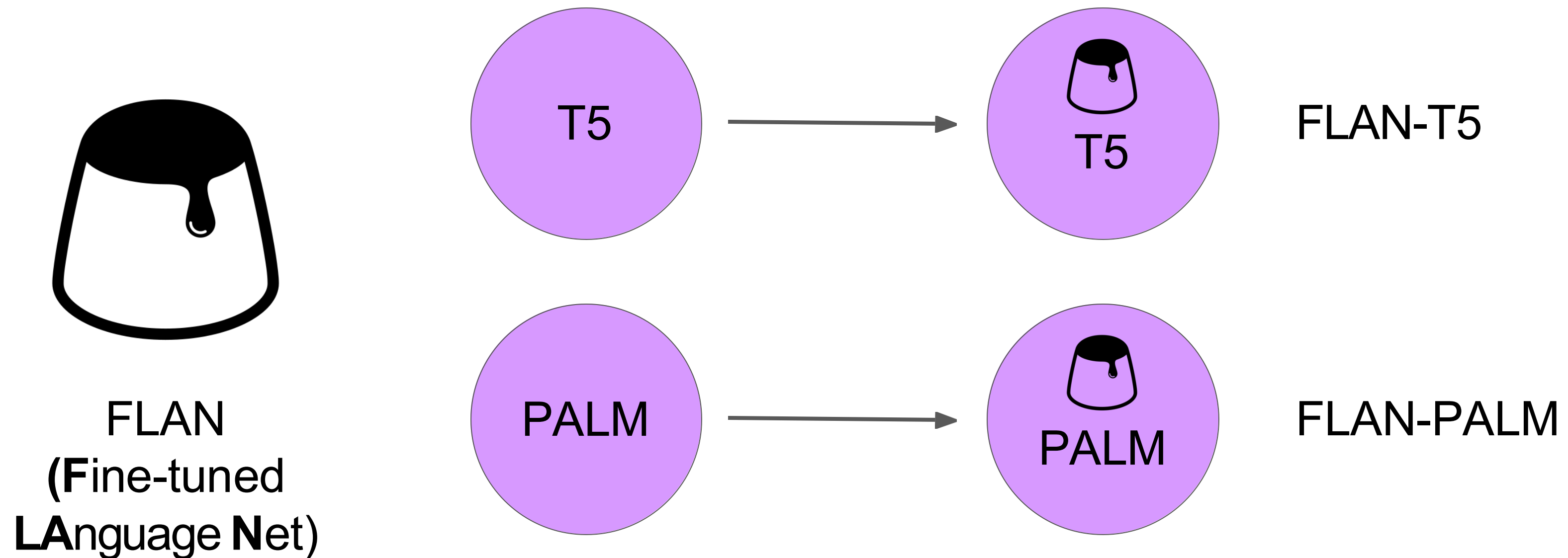


FLAN

“The metaphorical dessert to the main course of pretraining”

Instruction fine-tuning with FLAN

- FLAN models refer to a specific set of instructions used to perform instruction fine-tuning



FLAN-T5: Fine-tuned version of pre-trained T5 model

- FLAN-T5 is a great, general purpose, instruct model

T0-SF

- Commonsense Reasoning,
- Question Generation,
- Closed-book QA,
- Adversarial QA,
- Extractive QA

...

55 Datasets
14 Categories
193 Tasks

Muffin

- Natural language inference,
- Code instruction gen,
- Code repair
- Dialog context generation,
- Summarization (SAMSum)

...

69 Datasets
27 Categories
80 Tasks

CoT (reasoning)

- Arithmetic reasoning,
- Commonsense reasoning
- Explanation generation,
- Sentence composition,
- Implicit reasoning,

...

9 Datasets
1 Category
9 Tasks

Natural Instructions

- Cause effect classification,
- Commonsense reasoning,
- Named Entity Recognition,
- Toxic Language Detection,
- Question answering

...

372 Datasets
108 Categories
1554 Tasks

Source: Chung et al. 2022, "Scaling Instruction-Finetuned Language Models"

SAMSum: A dialogue dataset

Sample prompt training dataset (**samsum**) to fine-tune FLAN-T5 from pretrained T5

 Datasets: samsum		Tasks:  Summarization	Languages:  English
dialogue (string)		summary (string)	
"Amanda: I baked cookies. Do you want some? Jerry: Sure! Amanda: I'll bring you tomorrow :-)"		"Amanda baked cookies and will bring Jerry some tomorrow."	
"Olivia: Who are you voting for in this election? Oliver: Liberals as always. Olivia: Me too!! Oliver: Great"		"Olivia and Olivier are voting for liberals in this election. "	
"Tim: Hi, what's up? Kim: Bad mood tbh, I was going to do lots of stuff but ended up procrastinating Tim: What did..."		"Kim may try the pomodoro technique recommended by Tim to get more stuff done."	

Source: <https://huggingface.co/datasets/samsum>, <https://github.com/google-research/FLAN/blob/2c79a31/flan/v2/templates.py#L3285>

Sample FLAN-T5 prompt templates

"sumsum": [

("{dialogue}\n\nBriefly summarize that dialogue.", "{summary}"),

("Here is a dialogue:\n\n{dialogue}\n\nWrite a short summary!",

"{summary}"),

("Dialogue:\n\n{dialogue}\n\nWhat is a summary of this dialogue?",

"{summary}"),

("{dialogue}\n\nWhat was that dialogue about, in two sentences or less?",

"{summary}"),

("Here is a dialogue:\n\n{dialogue}\n\nWhat were they talking about?",

"{summary}"),

("Dialogue:\n\n{dialogue}\n\nWhat were the main points in that "

"conversation?", "{summary}"),

("Dialogue:\n\n{dialogue}\n\nWhat was going on in that conversation?"

"{summary}"),

]

Sample FLAN-T5 prompt templates

```
"samsun": [  
  ("{"dialogue"}\n\nBriefly summarize that dialogue.", "{"summary}"),  
  ("Here is a dialogue:\n{"dialogue"}\n\nWrite a shortsummary!",  
   "{"summary}"),  
  ("Dialogue:\n{"dialogue"}\n\nWhat is a summary of this dialogue?",  
   "{"summary}"),  
  ("{"dialogue"}\n\nWhat was that dialogue about, in two sentences or less?",  
   "{"summary}"),  
  ("Here is a dialogue:\n{"dialogue"}\n\nWhat were they talking about?",  
   "{"summary}"),  
  ("Dialogue:\n{"dialogue"}\n\nWhat were the main points in that "  
   "conversation?", "{"summary}"),  
  ("Dialogue:\n{"dialogue"}\n\nWhat was going on in that conversation?",  
   "{"summary}"),  
]
```

Improving FLAN-T5's summarization capabilities

 ShopBot

Hi there! How can I help you today?



I need to return a pair of jeans that I purchased



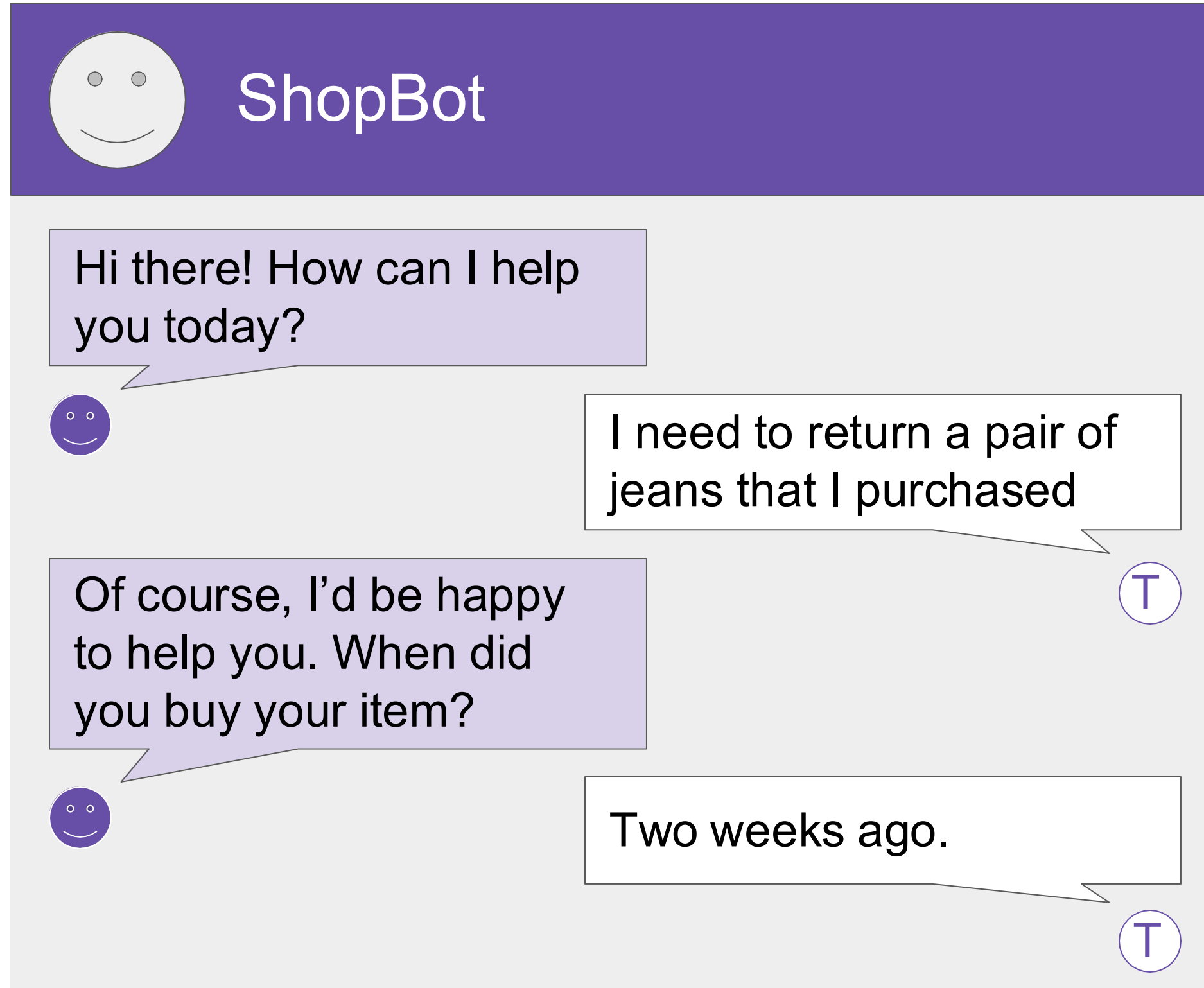
Of course, I'd be happy to help you. When did you buy your item?



Two weeks ago.



Improving FLAN-T5's summarization capabilities



The image shows a chat interface for a bot named "ShopBot". The header is a purple bar with a white smiley face icon and the text "ShopBot". The chat area has a light gray background. It contains three messages: a purple bubble from the bot saying "Hi there! How can I help you today?", a white bubble from the user saying "I need to return a pair of jeans that I purchased" with a small purple smiley face icon to its left and a purple circle with a white 'T' to its right, and another purple bubble from the bot saying "Of course, I'd be happy to help you. When did you buy your item?". Below this is another white bubble from the user saying "Two weeks ago." with a small purple smiley face icon to its left and a purple circle with a white 'T' to its right.

ShopBot

Hi there! How can I help you today?

I need to return a pair of jeans that I purchased

Of course, I'd be happy to help you. When did you buy your item?

Two weeks ago.

Goal: Summarize conversations to identify actions to take

Improving FLAN-T5's summarization capabilities

Further fine-tune FLAN-T5 with a domain-specific instruction dataset (**dialogsum**)

Datasets: knkarthick/ dialogsum like 13		
Tasks:	Summarization	Text2Text Generation
Languages:	English	Multilinguality: monolingual
Language Creators:	expert-generated	Annotations Creators: expert-generated
Source Datasets:	original	License: mit
Dataset card Files and versions Community 4		
Dataset Preview		
Split		
train (12.5k rows)		
id (string)	dialogue (string)	summary (string)
"train_0"	"#Person1#: Hi, Mr. Smith. I'm Doctor Hawkins. Why are you here today? #Person2#: I found it would be a good..."	"Mr. Smith's getting a check-up, and Doctor Hawkins advises him to have one every year. Hawkins'll give some..."
"train_1"	"#Person1#: Hello Mrs. Parker, how have you been? #Person2#: Hello Dr. Peters. Just fine thank you. Ricky..."	"Mrs Parker takes Ricky for his vaccines. Dr. Peters checks the record and then gives Ricky a vaccine."
"train_2"	"#Person1#: Excuse me, did you see a set of keys? #Person2#: What kind of keys? #Person1#: Five keys and a small foot ornament. #Person2#: What a shame! I didn't see them. #Person1#: Well, can you help me look for it? That's my first time here. #Person2#: Sure. It's my pleasure. I'd like to help you look for the missing keys. #Person1#: It's very kind of you. #Person2#: It's not a big deal. Hey, I found them. #Person1#: Oh, thank God! I don't know how to thank you, guys. #Person2#: You're welcome."	"#Person1#'s looking for a set of keys and asks for #Person2#'s help to find them."

Example support-dialog summarization

Prompt (created from template)

Summarize the following conversation.

Tommy: Hello. My name is Tommy Sandals, I have a reservation.

Mike: May I see some identification, sir, please?

Tommy: Sure. Here you go.

Mike: Thank you so much. Have you got a credit card, Mr. Sandals?

Tommy: I sure do.

Mike: Thank you, sir. You'll be in room 507, nonsmoking, queen bed.

Tommy: That's great, thank you!

Mike: Enjoy your stay!

Source: <https://huggingface.co/datasets/knkarthick/dialogsum/viewer/knkarthick--dialogsum/>

Summary before fine-tuning FLAN-T5 with our dataset

Prompt (created from template)

Summarize the following conversation.

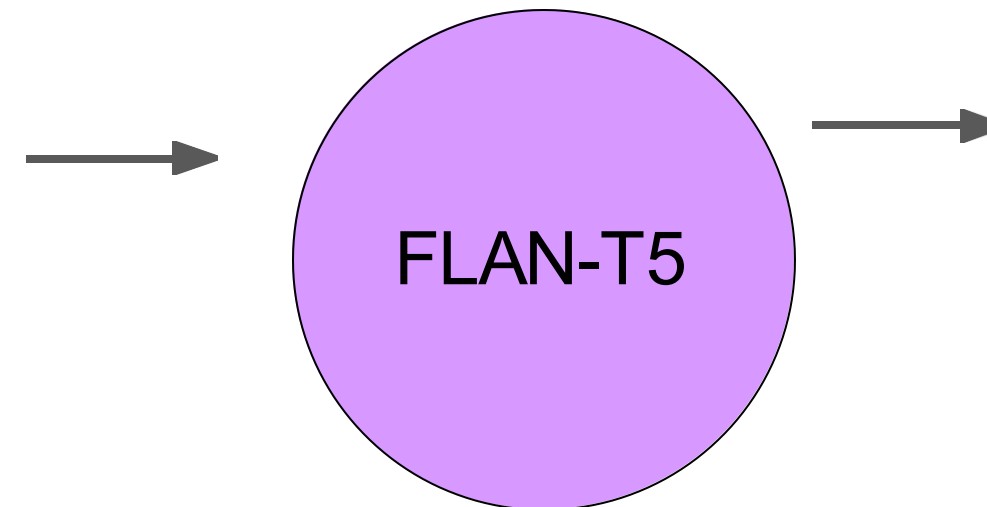
Tommy: Hello. My name is Tommy Sandals, I have a reservation.

Mike: May I see some
...
...
...

Tommy: That's great, thank you!

Mike: Enjoy your stay!

Model



Completion (Summary)

Tommy Sandals has a reservation for a room at the Venetian Hotel in Las Vegas.

Adequate completion, but does not match human baseline.

Human baseline summary:
Tommy Sandals has got a reservation. Mike asks for his identification and credit card and helps his check-in.

Summary before fine-tuning FLAN-T5 with our dataset

Prompt (created from template)

Summarize the following conversation.

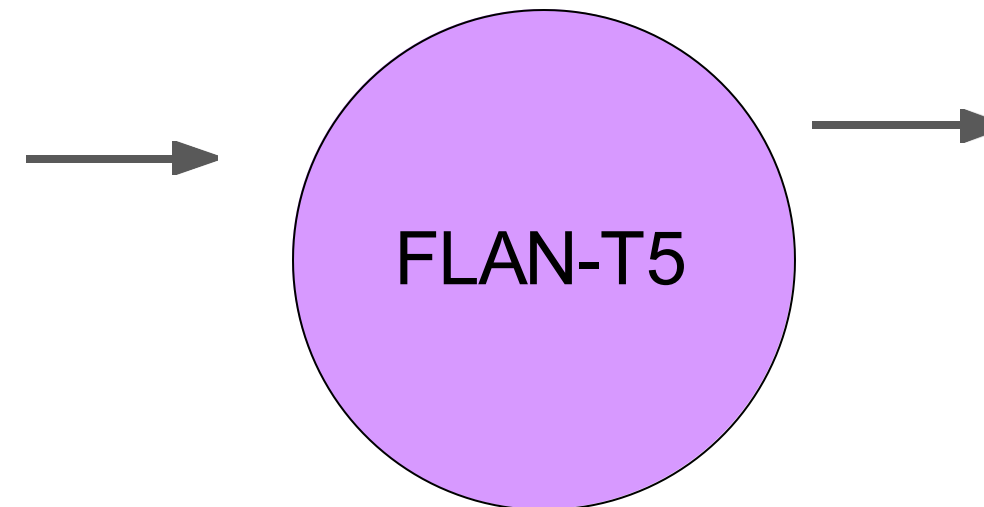
Tommy: Hello. My name is Tommy Sandals, I have a reservation.

Mike: May I see some
...
...
...

Tommy: That's great, thank you!

Mike: Enjoy your stay!

Model



Completion (Summary)

Tommy Sandals has a reservation for a room at the **Venetian** Hotel in Las Vegas.

Adequate completion, but does not match human baseline.

Human baseline summary:
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Summary before fine-tuning FLAN-T5 with our dataset

Prompt (created from template)

Summarize the following conversation.

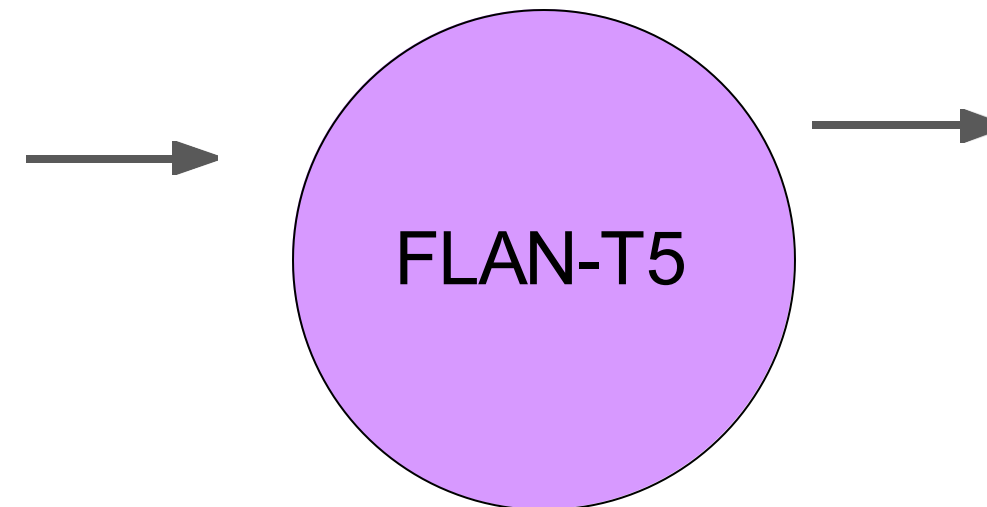
Tommy: Hello. My name is Tommy Sandals, I have a reservation.

Mike: May I see some
...
...
...

Tommy: That's great, thank you!

Mike: Enjoy your stay!

Model



Completion (Summary)

Tommy Sandals has a reservation for a room at the Venetian Hotel in Las Vegas.

Adequate completion, but does not match human baseline.

Human baseline summary:
Tommy Sandals has got a reservation. Mike asks for his identification and credit card and helps his check-in.

Summary **after** fine-tuning FLAN-T5 with our dataset

Prompt (created from template)

Summarize the following conversation.

Tommy: Hello. My name is Tommy Sandals, I have a reservation.

Mike: May I see some

...

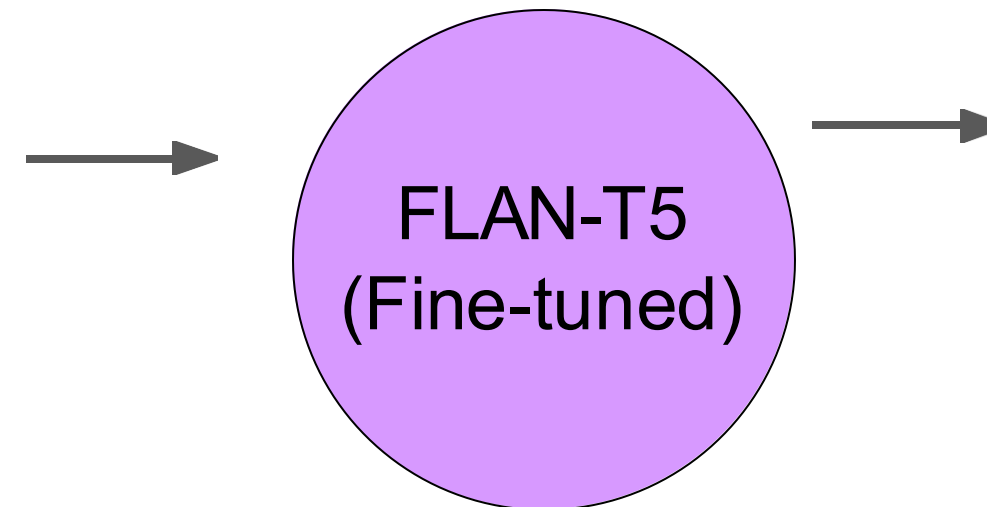
...

...

Tommy: That's great, thank you!

Mike: Enjoy your stay!

Model



Completion (Summary)

Tommy Sandals has a reservation and checks in showing his ID and credit card. Mike helps him to check in and approves his reservation.

*Better summary,
more-closely matches
human baseline.*

Fine-tuning with your own data

 ShopBot

Hi there! How can I help you today?



I need to return a pair of jeans that I purchased



Of course, I'd be happy to help you. When did you buy your item?



Two weeks ago.



Model evaluation metrics

LLM Evaluation - Challenges

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

LLM Evaluation - Challenges

“Mike really loves drinking tea.”



=

“Mike adores sipping tea.”



“Mike does not drink coffee.”

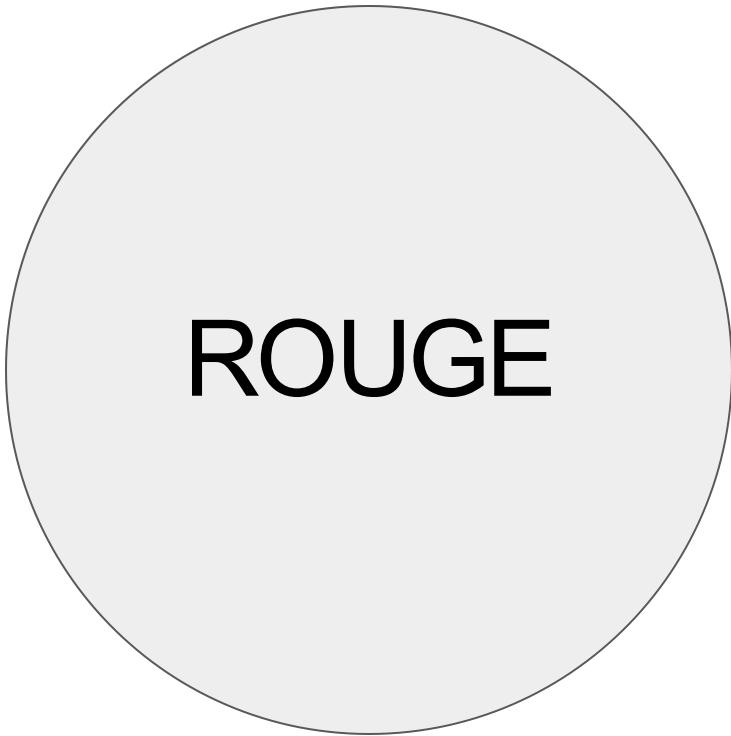


≠

“Mike does drink coffee.”



LLM Evaluation - Metrics



ROUGE

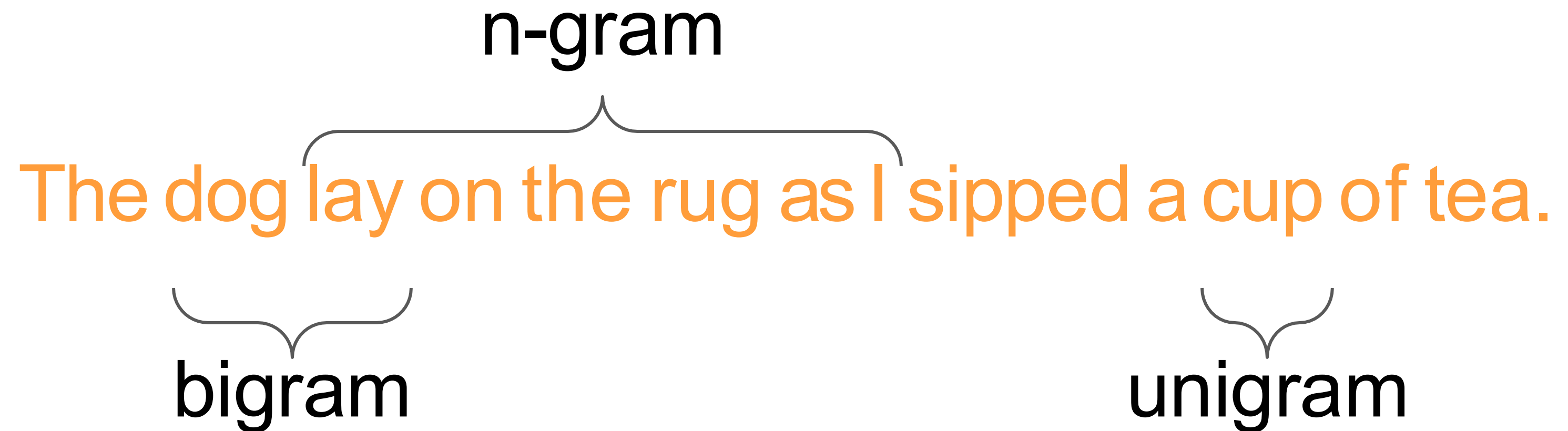
- Used for text summarization
- Compares a summary to one or more reference summaries



BLEU
SCORE

- Used for text translation
- Compares to human-generated translations

LLM Evaluation - Metrics - Terminology



LLM Evaluation - Metrics - ROUGE-1

Reference (human):

It is cold outside.

Generated output:

It is very cold outside.

$$\text{ROUGE-1 Recall} = \frac{\text{unigram matches}}{\text{unigrams in reference}} = \frac{4}{4} = 1.0$$

$$\text{ROUGE-1 Precision:} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{5} = 0.8$$

$$\text{ROUGE-1 F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.8}{1.8} = 0.89$$

LLM Evaluation - Metrics - ROUGE-1

Reference (human):

It is cold outside.

Generated output:

It is not cold outside.

$$\text{ROUGE-1 Recall} = \frac{\text{unigram matches}}{\text{unigrams in reference}} = \frac{4}{4} = 1.0$$

$$\text{ROUGE-1 Precision:} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{5} = 0.8$$

$$\text{ROUGE-1 F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.8}{1.8} = 0.89$$

LLM Evaluation - Metrics - ROUGE-2

Reference (human):

It is cold outside.

It is

is cold

cold outside

Generated output:

It is very cold outside.

It is

is very

very cold

cold outside

LLM Evaluation - Metrics - ROUGE-2

Reference (human):

It is cold outside.

It is

is cold

cold outside

Generated output:

It is very cold outside.

It is

is very

very cold

cold outside

$$\text{ROUGE-2 Recall:} = \frac{\text{bigram matches}}{\text{bigrams in reference}} = \frac{2}{3} = 0.67$$

$$\text{ROUGE-2 Precision:} = \frac{\text{bigram matches}}{\text{bigrams in output}} = \frac{2}{4} = 0.5$$

$$\text{ROUGE-2 F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.335}{1.17} = 0.57$$

LLM Evaluation - Metrics - ROUGE-L

Reference (human):

It is cold outside.

Generated output:

It is very cold outside.

Longest common subsequence (LCS):

It is

cold outside

2

LLM Evaluation - Metrics - ROUGE-L

Reference (human):

It is cold outside.

Generated output:

It is very cold outside.

$$\text{ROUGE-L Recall:} = \frac{\text{LCS}(\text{Gen, Ref})}{\text{unigrams in reference}} = \frac{2}{4} = 0.5$$

$$\text{ROUGE-L Precision:} = \frac{\text{LCS}(\text{Gen, Ref})}{\text{unigrams in output}} = \frac{2}{5} = 0.4$$

$$\text{ROUGE-L F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.2}{0.9} = 0.44$$

LLM Evaluation - Metrics - ROUGE-L

Reference (human):

It is cold outside.

Generated output:

It is very cold outside.

LCS:

Longest common subsequence

$$\text{ROUGE-L Recall:} = \frac{\text{LCS}(\text{Gen}, \text{Ref})}{\text{unigrams in reference}} = \frac{2}{4} = 0.5$$

$$\text{ROUGE-L Precision:} = \frac{\text{LCS}(\text{Gen}, \text{Ref})}{\text{unigrams in output}} = \frac{2}{5} = 0.4$$

$$\text{ROUGE-L F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.2}{0.9} = 0.44$$

LLM Evaluation - Metrics - ROUGE hacking

Reference (human):

It is cold outside.

Generated output:

cold cold cold cold

LLM Evaluation - Metrics - ROUGE clipping

Reference (human):

It is cold outside.

Generated output:

cold cold cold cold

$$\text{ROUGE-1 Precision} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{4} = 1.0$$

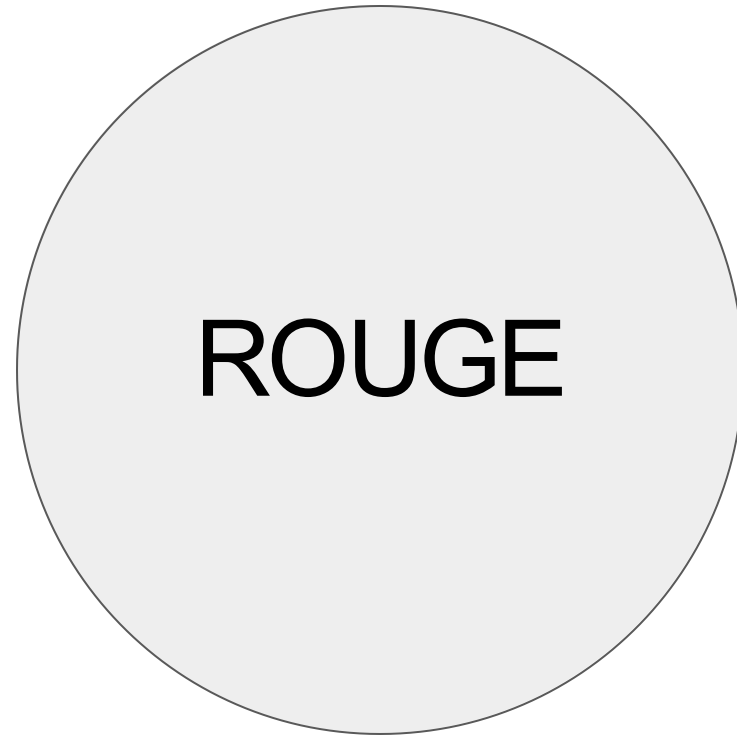

$$\text{Modified precision} = \frac{\text{clip(unigram matches)}}{\text{unigrams in output}} = \frac{1}{4} = 0.25$$

Generated output:

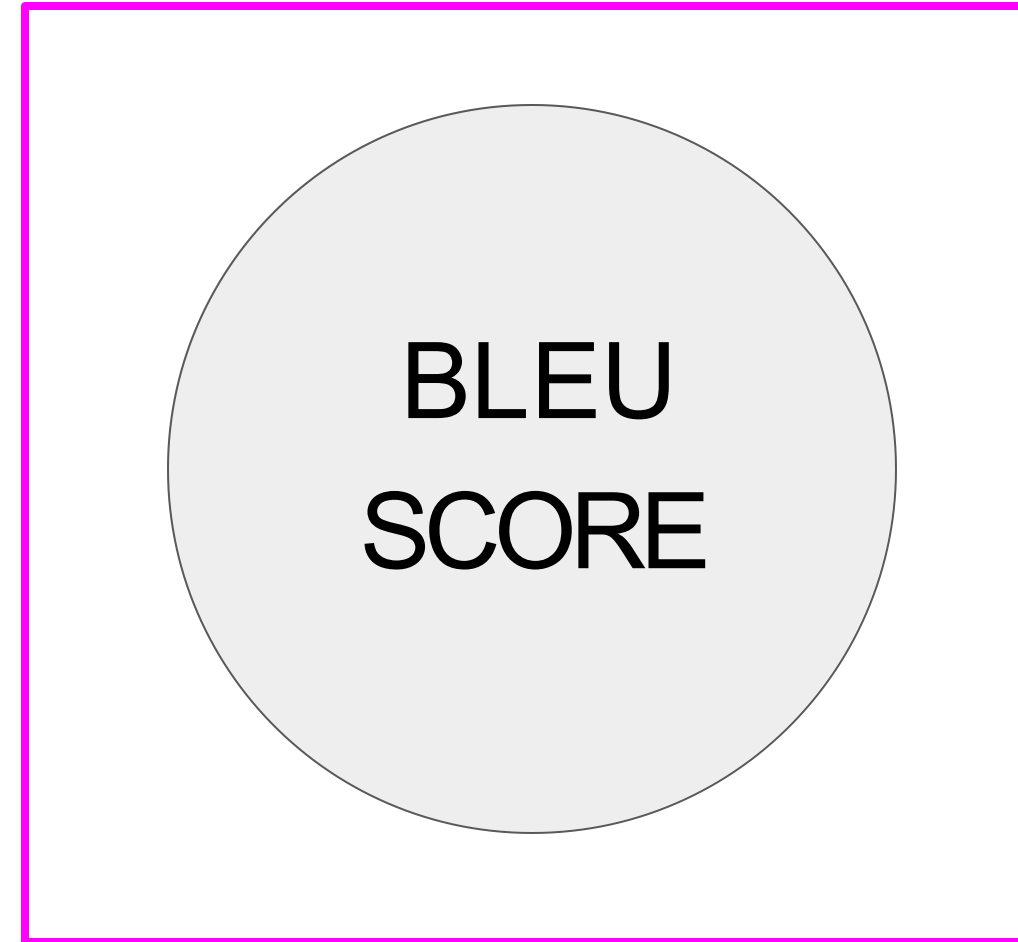
outside cold it is

$$\text{Modified precision} = \frac{\text{clip(unigram matches)}}{\text{unigrams in output}} = \frac{4}{4} = 1.0$$


LLM Evaluation - Metrics



- Used for text summarization
- Compares a summary to one or more reference summaries



- Used for text translation
- Compares to human-generated translations

LLM Evaluation - Metrics - BLEU

BLEU metric = Avg(precision across range of n-gram sizes)

Reference (human):

I am very happy to say that I am drinking a warm cup of tea.

Generated output:

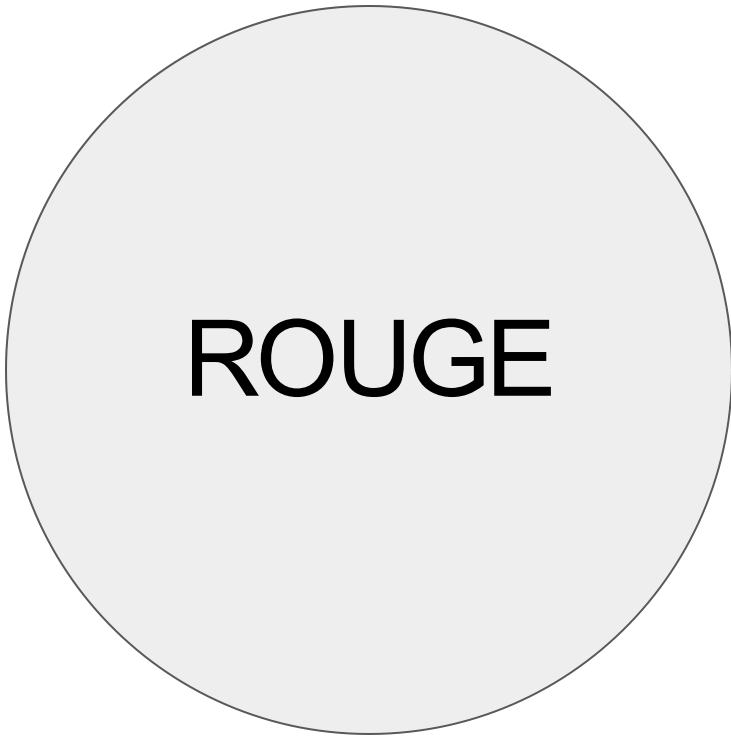
I am very happy that I am drinking a cup of tea. - BLEU 0.495

I am very happy that I am drinking a warm cup of tea. - BLEU 0.730

I am very happy to say that I am drinking a warm tea. - BLEU 0.798

I am very happy to say that I am drinking a warm cup of tea. - BLEU 1.000

LLM Evaluation - Metrics



ROUGE

- Used for text summarization
- Compares a summary to one or more reference summaries



BLEU
SCORE

- Used for text translation
- Compares to human-generated translations

Benchmarks

Evaluation benchmarks



MMLU (Massive Multitask
Language Understanding)

BIG-bench 

GLUE



The tasks included in SuperGLUE benchmark:

Corpus	Train	Test	Task	Metrics	Domain
Single-Sentence Tasks					
CoLA	8.5k	1k	acceptability	Matthews corr.	misc.
SST-2	67k	1.8k	sentiment	acc.	movie reviews
Similarity and Paraphrase Tasks					
MRPC	3.7k	1.7k	paraphrase	acc./F1	news
STS-B	7k	1.4k	sentence similarity	Pearson/Spearman corr.	misc.
QQP	364k	391k	paraphrase	acc./F1	social QA questions
Inference Tasks					
MNLI	393k	20k	NLI	matched acc./mismatched acc.	misc.
QNLI	105k	5.4k	QA/NLI	acc.	Wikipedia
RTE	2.5k	3k	NLI	acc.	news, Wikipedia
WNLI	634	146	coreference/NLI	acc.	fiction books

Source: Wang et al. 2018, “GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding”

SuperGLUE



The tasks included in SuperGLUE benchmark:

Corpus	Train	Dev	Test	Task	Metrics	Text Sources
BoolQ	9427	3270	3245	QA	acc.	Google queries, Wikipedia
CB	250	57	250	NLI	acc./F1	various
COPA	400	100	500	QA	acc.	blogs, photography encyclopedia
MultiRC	5100	953	1800	QA	F1 _a /EM	various
ReCoRD	101k	10k	10k	QA	F1/EM	news (CNN, Daily Mail)
RTE	2500	278	300	NLI	acc.	news, Wikipedia
WiC	6000	638	1400	WSD	acc.	WordNet, VerbNet, Wiktionary
WSC	554	104	146	coref.	acc.	fiction books

Source: Wang et al. 2019, “SuperGLUE: A Stickier Benchmark for General-Purpose Language Understanding Systems”

GLUE and SuperGLUE leaderboards

GLUE

SuperGLUE

Paper

Code

Tasks

Leaderboard

FAQ

Diagnostics

Submit

Login

SuperGLUE

GLUE

Rank Name

1 Microsoft Alexander v-team

2 JDExplore d-team

3 Microsoft Alexander v-team

4 DTRL Team

5 ERNIE Team - Baidu

6 AliceMind & DTRL

7 DeBERTa Team - Microsoft

8 HFL iFLYTEK

9 PING-AN Omni-Sight

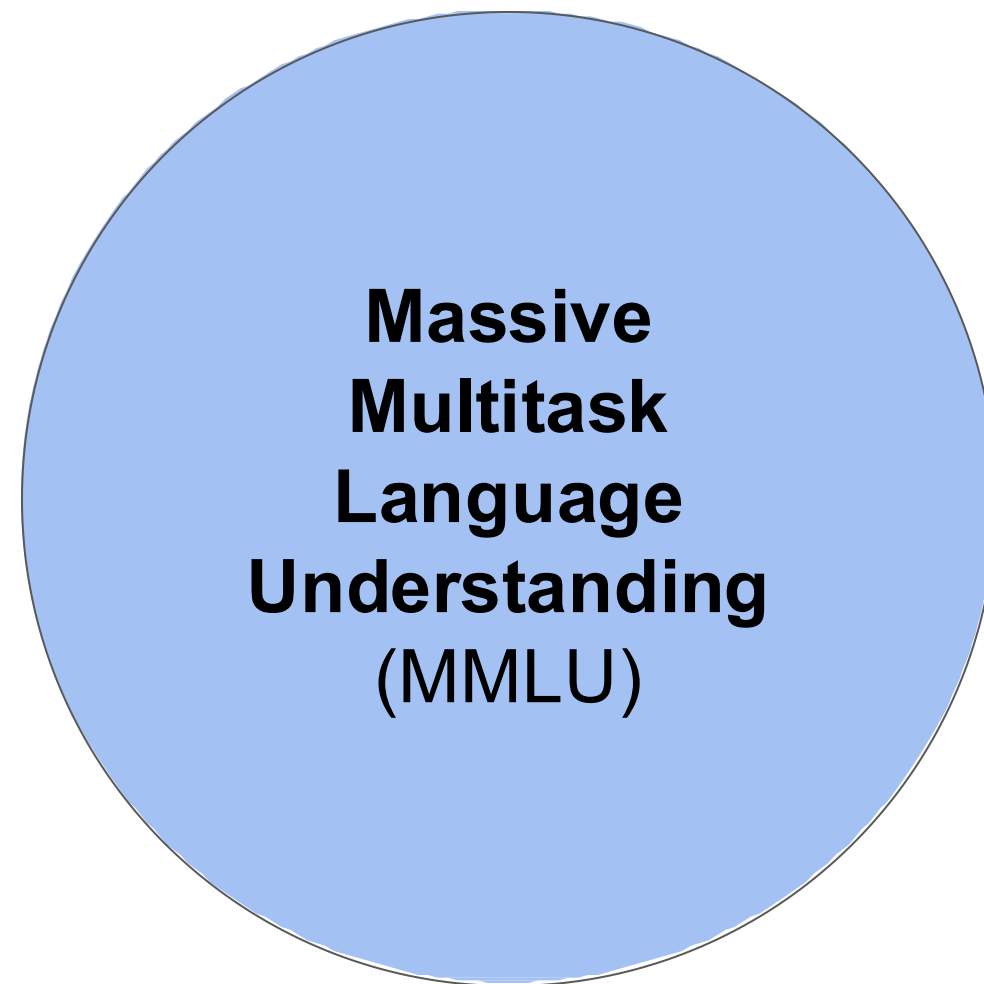
10 T5 Team - Google

Leaderboard Version: 2.0

Rank	Name	Model	URL	Score	BoolQ	CB	COPA	MultiRC	ReCoRD	RTE	WIC	WSC	AX-b	AX-g	
1	JDExplore d-team	Vega v2		91.3	90.5	98.6/99.2	99.4	88.2/62.4	94.4/93.9	96.0	77.4	98.6	-0.4	100.0/50.0	
+	2	Liam Fedus	ST-MoE-32B		91.2	92.4	96.9/98.0	99.2	89.6/65.8	95.1/94.4	93.5	77.7	96.6	72.3	96.1/94.1
	3	Microsoft Alexander v-team	Turing NLR v5		90.9	92.0	95.9/97.6	98.2	88.4/63.0	96.4/95.9	94.1	77.1	97.3	67.8	93.3/95.5
	4	ERNIE Team - Baidu	ERNIE 3.0		90.6	91.0	98.6/99.2	97.4	88.6/63.2	94.7/94.2	92.6	77.4	97.3	68.6	92.7/94.7
	5	Yi Tay	PaLM 540B		90.4	91.9	94.4/96.0	99.0	88.7/63.6	94.2/93.3	94.1	77.4	95.9	72.9	95.5/90.4
+	6	Zirui Wang	T5 + UDG, Single Model (Google Brain)		90.4	91.4	95.8/97.6	98.0	88.3/63.0	94.2/93.5	93.0	77.9	96.6	69.1	92.7/91.9
+	7	DeBERTa Team - Microsoft	DeBERTa / TuringNLRv4		90.3	90.4	95.7/97.6	98.4	88.2/63.7	94.5/94.1	93.2	77.5	95.9	66.7	93.3/93.8

Disclaimer: metrics may not be up-to-date. Check <https://super.gluebenchmark.com> and <https://gluebenchmark.com/leaderboard> for the latest.

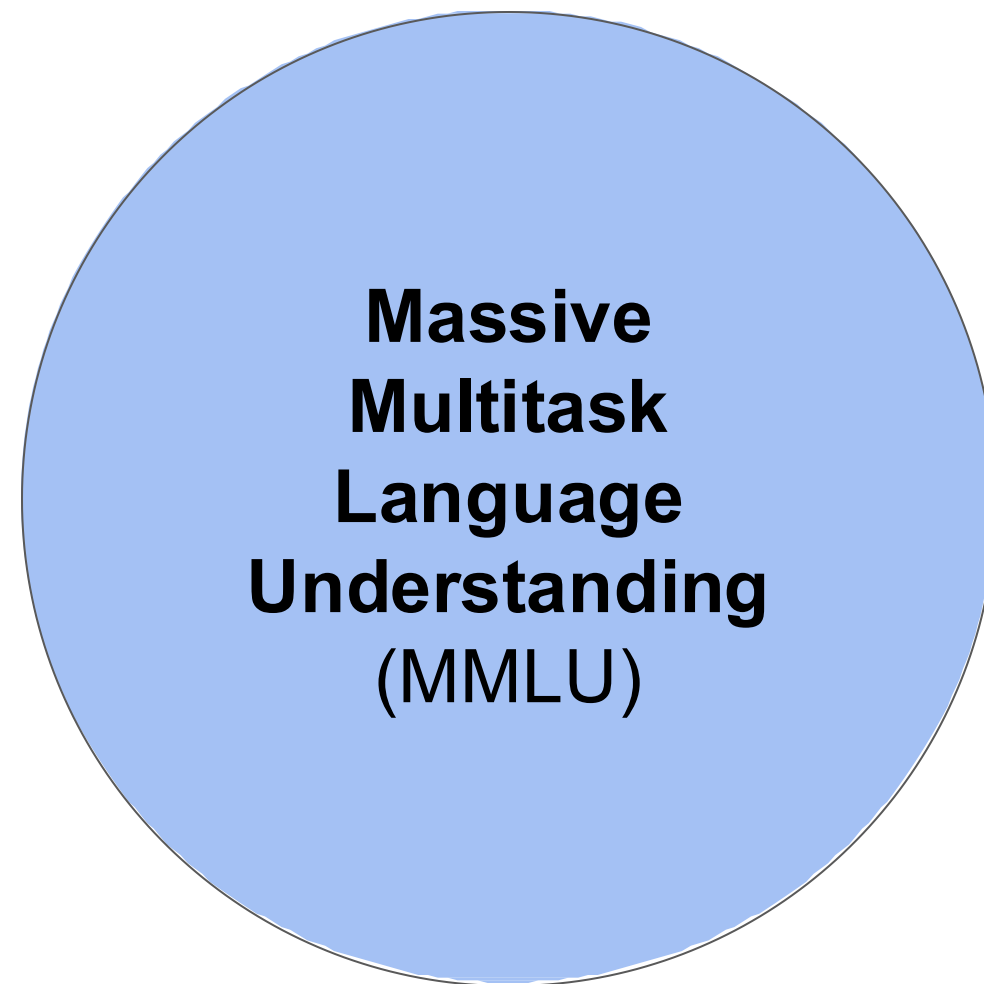
Benchmarks for massive models



2021

Source: Hendrycks, 2021. "Measuring Massive Multitask Language Understanding"

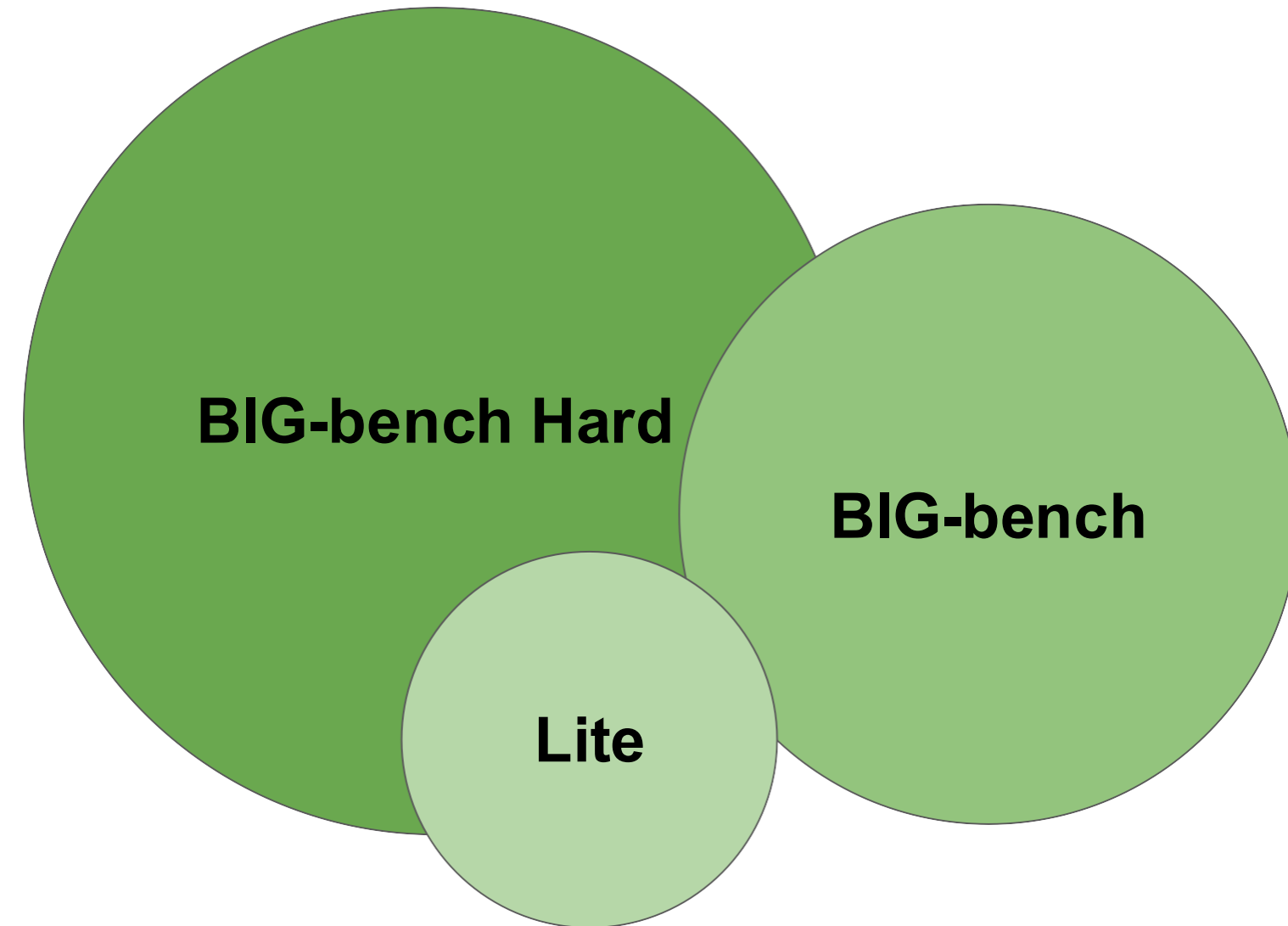
Benchmarks for massive models



**Massive
Multitask
Language
Understanding
(MMLU)**

2021

Source: Hendrycks, 2021. "Measuring Massive Multitask Language Understanding"



BIG-bench Hard

BIG-bench

Lite

2022

Source: Suzgun et al. 2022. "Challenging BIG-Bench tasks and whether chain-of-thought can solve them"

Holistic Evaluation of Language Models (HELM)



Metrics:

1. Accuracy
2. Calibration
3. Robustness
4. Fairness
5. Bias
6. Toxicity
7. Efficiency

Scenarios

Models

	J1-Jumbo	J1-Grande	J1-Large	Anthropic-LM	BLOOM	T0pp
NaturalQuestions (open)			✓	✓	✓	✓
NaturalQuestions (closed)	✓	✓	✓	✓	✓	✓
BoolQ	✓	✓	✓	✓	✓	✓
NarrativeQA	✓	✓	✓	✓	✓	✓
QuAC	✓	✓	✓	✓	✓	✓
HellaSwag	✓	✓	✓	✓	✓	✓
OpenBookQA	✓	✓	✓	✓	✓	✓
TruthfulQA	✓	✓	✓	✓	✓	✓
MMLU	✓	✓	✓	✓	✓	✓
MS MARCO				✓	✓	
TREC				✓	✓	
XSUM	✓	✓	✓	✓	✓	✓
CNN/DM	✓	✓	✓	✓	✓	✓
IMDB	✓	✓	✓	✓	✓	✓
CivilComments	✓	✓	✓	✓	✓	✓
RAFT	✓	✓	✓	✓	✓	✓

Holistic Evaluation of Language Models (HELM)



Center for Research on Foundation Models

HELM

Models

Scenarios

Results

Raw runs

v0.2.2 (last updated 2023-03-19)

Core scenarios

The scenarios where we evaluate all the models.

[\[Accuracy](#)
[| Calibration](#)
[| Robustness](#)
[| Fairness](#)
[| Efficiency](#)
[| General information](#)
[| Bias](#)
[| Toxicity](#)
[| Summarization metrics](#)
[| JSON \]](#)

Accuracy

Model/adaptor	Mean win rate ↑ [sort]	MMLU - EM ↑ [sort]	BoolQ - EM ↑ [sort]	NarrativeQA - F1 ↑ [sort]	NaturalQuestions (closed-book) - F1 ↑ [sort]	NaturalQuestions (open-book) - F1 ↑ [sort]	QuAC - F1 ↑ [sort]	HellaSwag - EM ↑ [sort]	OpenbookQA - EM ↑ [sort]	TruthfulQA - EM ↑ [sort]
Cohere Command beta (52.4B)	0.93	0.452	0.856	0.752	0.372	0.76	0.432	0.811	0.582	0.269
text-davinci-002	0.93	0.568	0.877	0.727	0.383	0.713	0.445	0.815	0.594	0.61
text-davinci-	0.898	0.569	0.881	0.727	0.406	0.77	0.525	0.822	0.646	0.593

Disclaimer: metrics may not be up-to-date. Check <https://crfm.stanford.edu/helm/latest> for the latest.

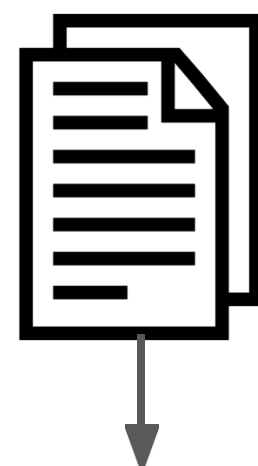
Key takeaways



LLM fine-tuning process

LLM fine-tuning

Training dataset

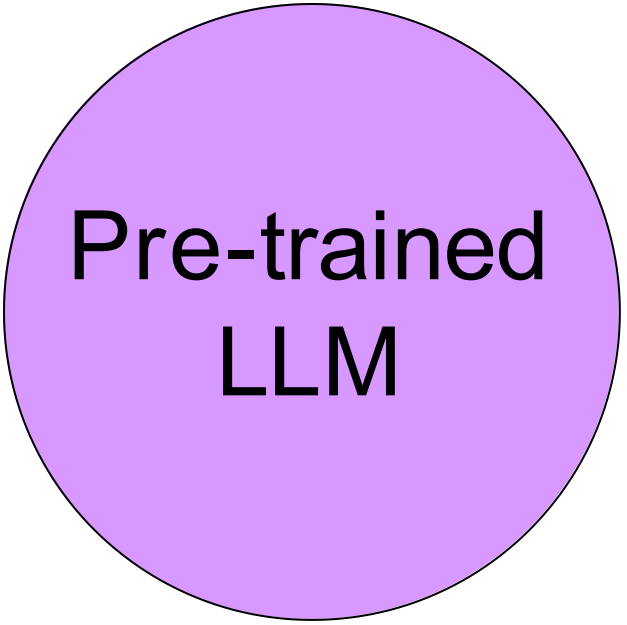


Prompt:

Classify this review:
I loved this DVD!

Sentiment:

Model



LLM completion:

Label:

Loss: Cross

LLM fine-tuning process

LLM fine-tuning

Training dataset



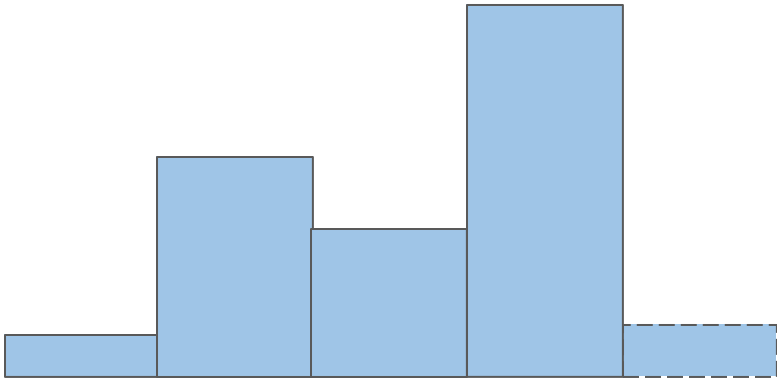
Prompt:

Classify this review:
I loved this DVD!
Sentiment:

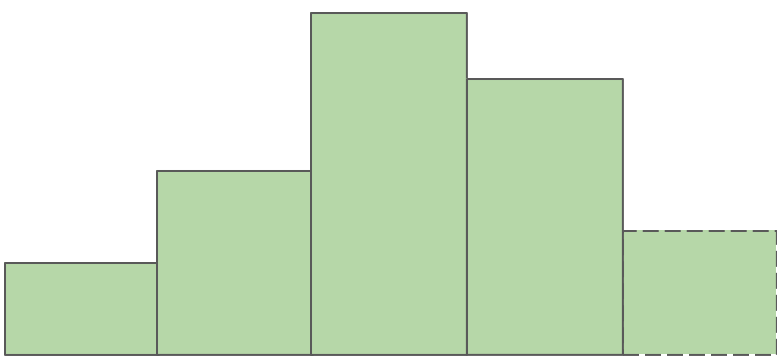
Model



LLM completion:



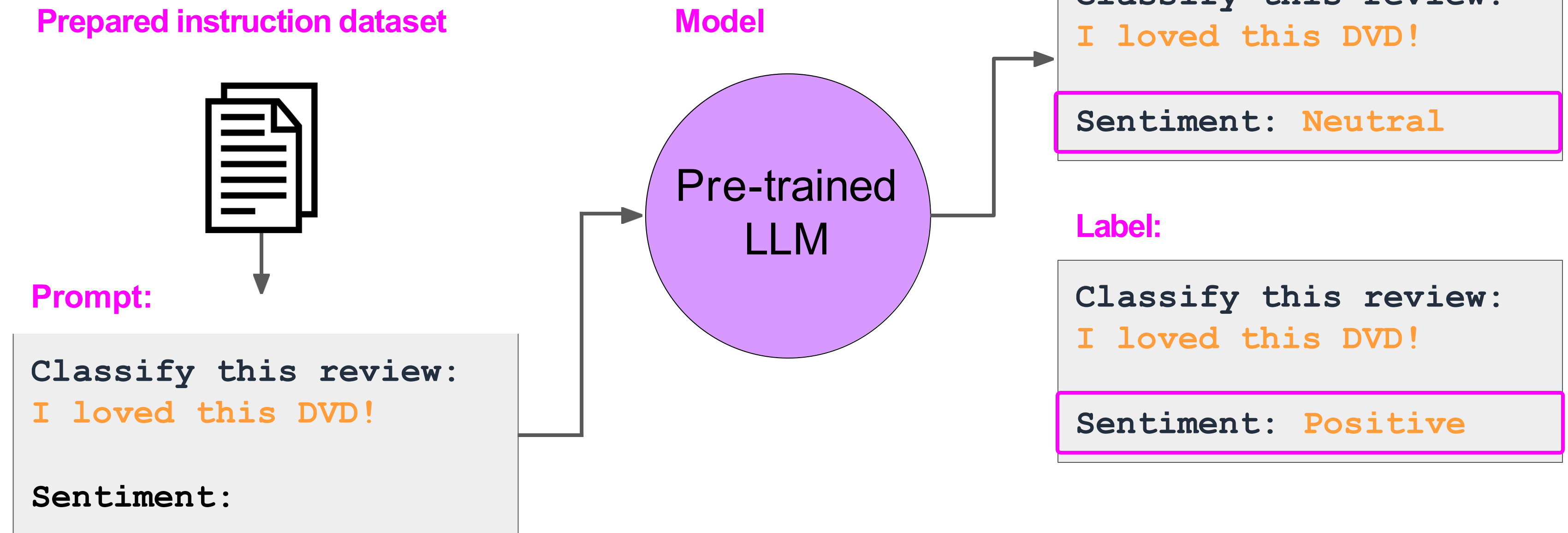
Label:



Loss: Cross-Entropy

LLM fine-tuning process

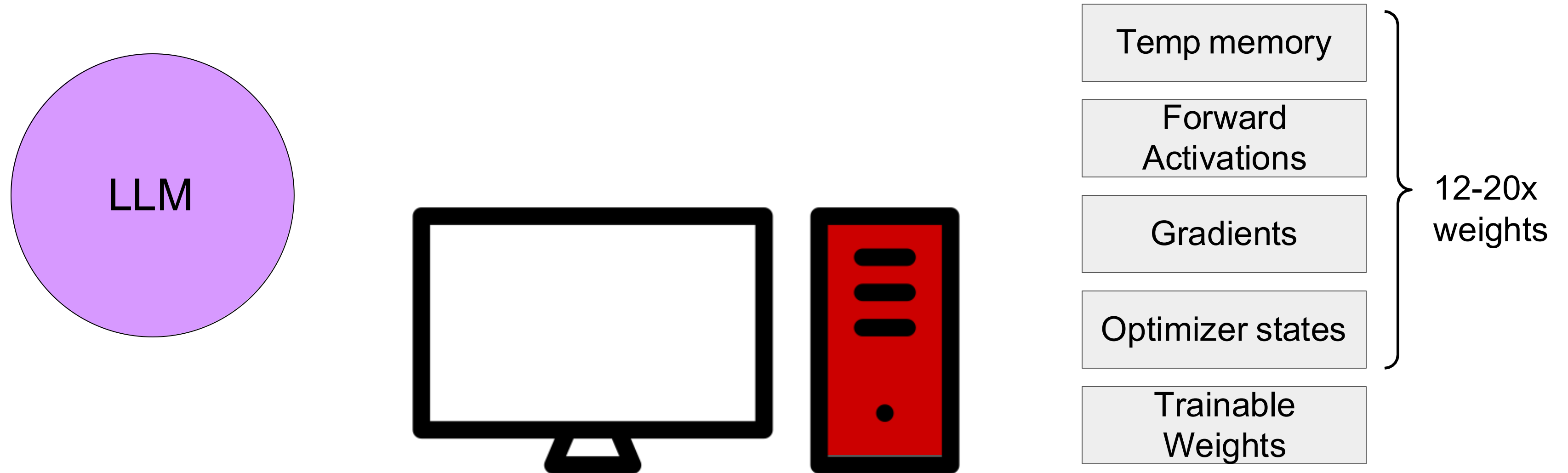
LLM fine-tuning



Parameter- efficient Fine-tuning (PEFT)

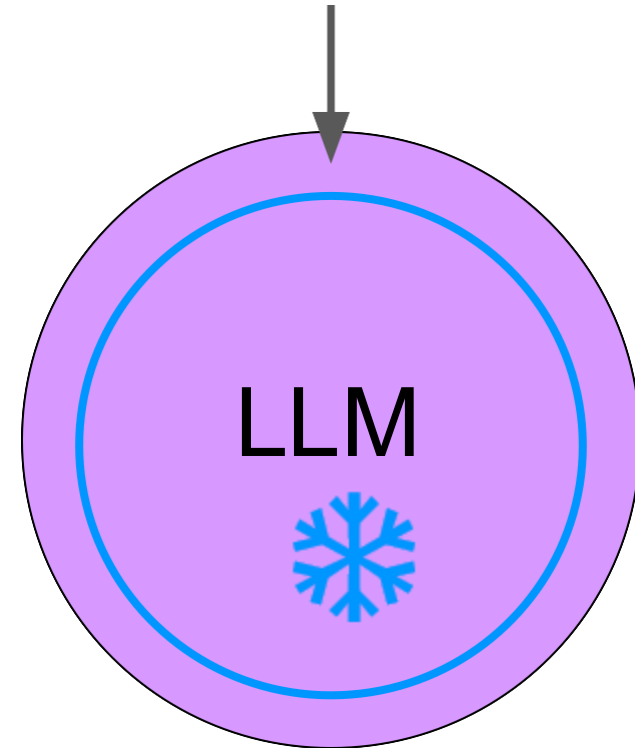


Full fine-tuning of large LLMs is challenging

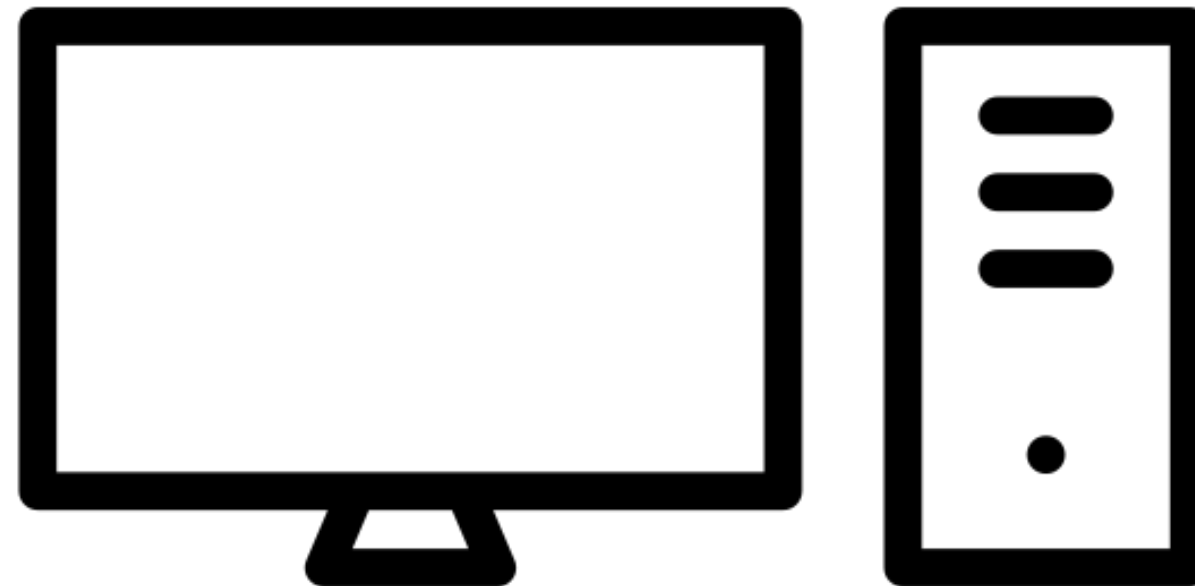


Parameter efficient fine-tuning (PEFT)

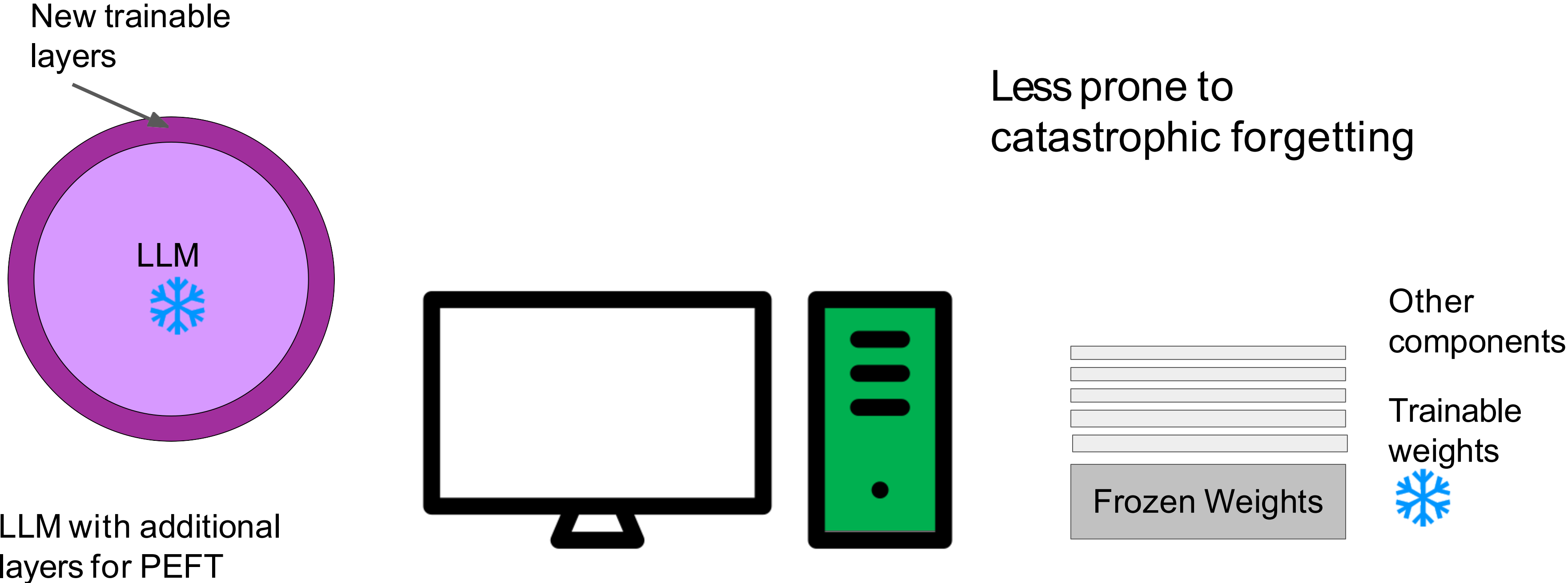
Small number of trainable layers



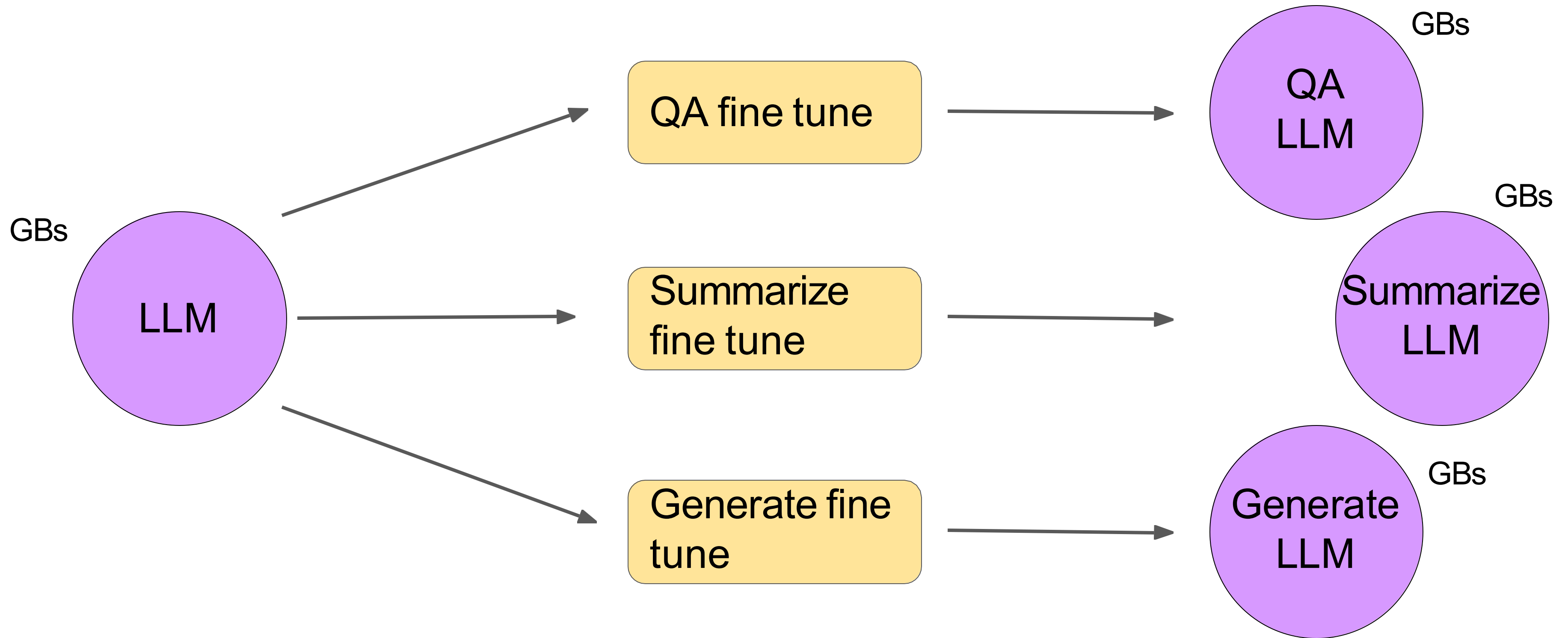
LLM with most layers frozen



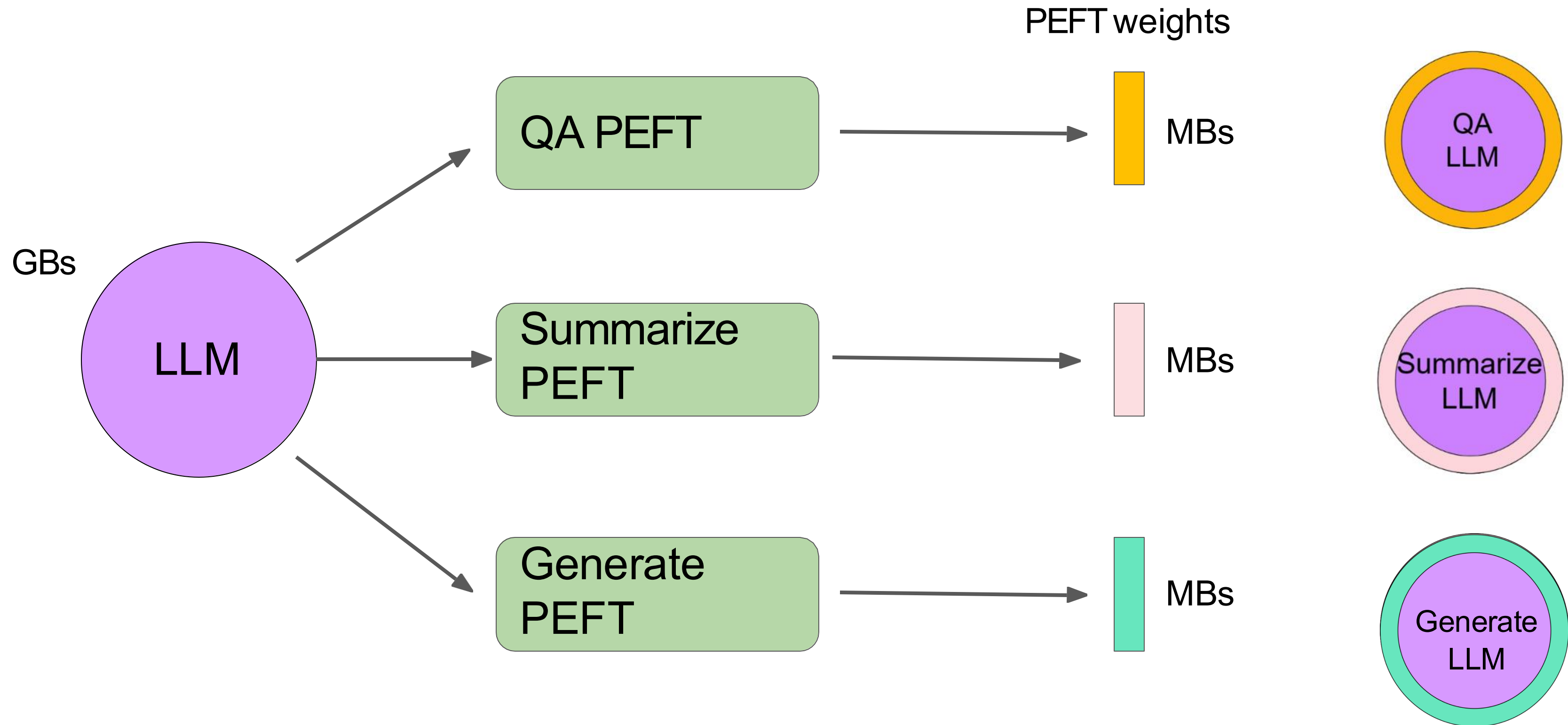
Parameter efficient fine-tuning (PEFT)



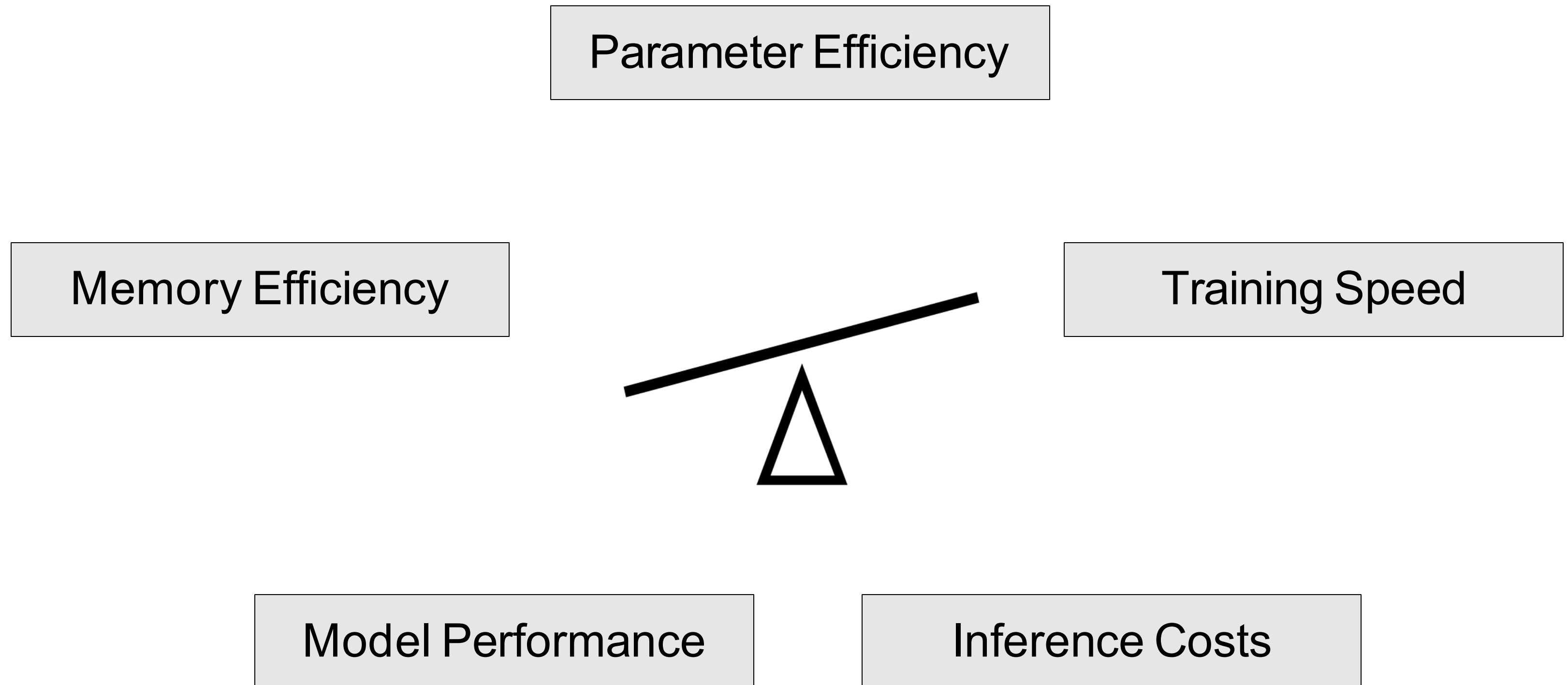
Full fine-tuning creates full copy of original LLM per task



PEFT fine-tuning saves space and is flexible



PEFT Trade-offs



PEFT methods

Selective

Select subset of initial LLM parameters to fine-tune

Reparameterization

Reparameterize model weights using a low-rank representation

Source: Lialin et al. 2023, “Scaling Down to Scale Up: A Guide to Parameter-Efficient Fine-Tuning”,

PEFT methods

Selective

Select subset of initial LLM parameters to fine-tune

Reparameterization

Reparameterize model weights using a low-rank representation

LoRA

Additive

Add trainable layers or parameters to model

Adapters

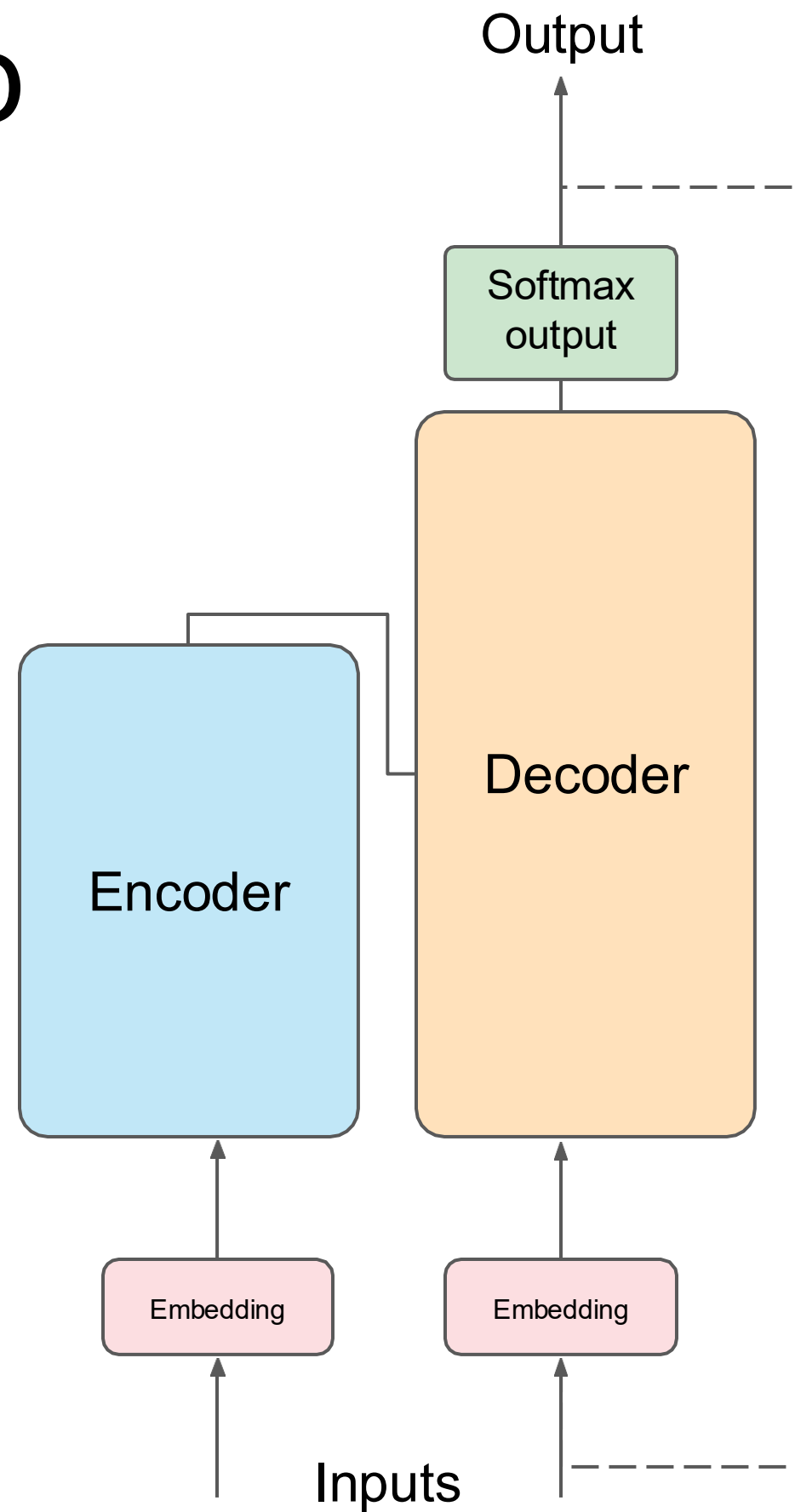
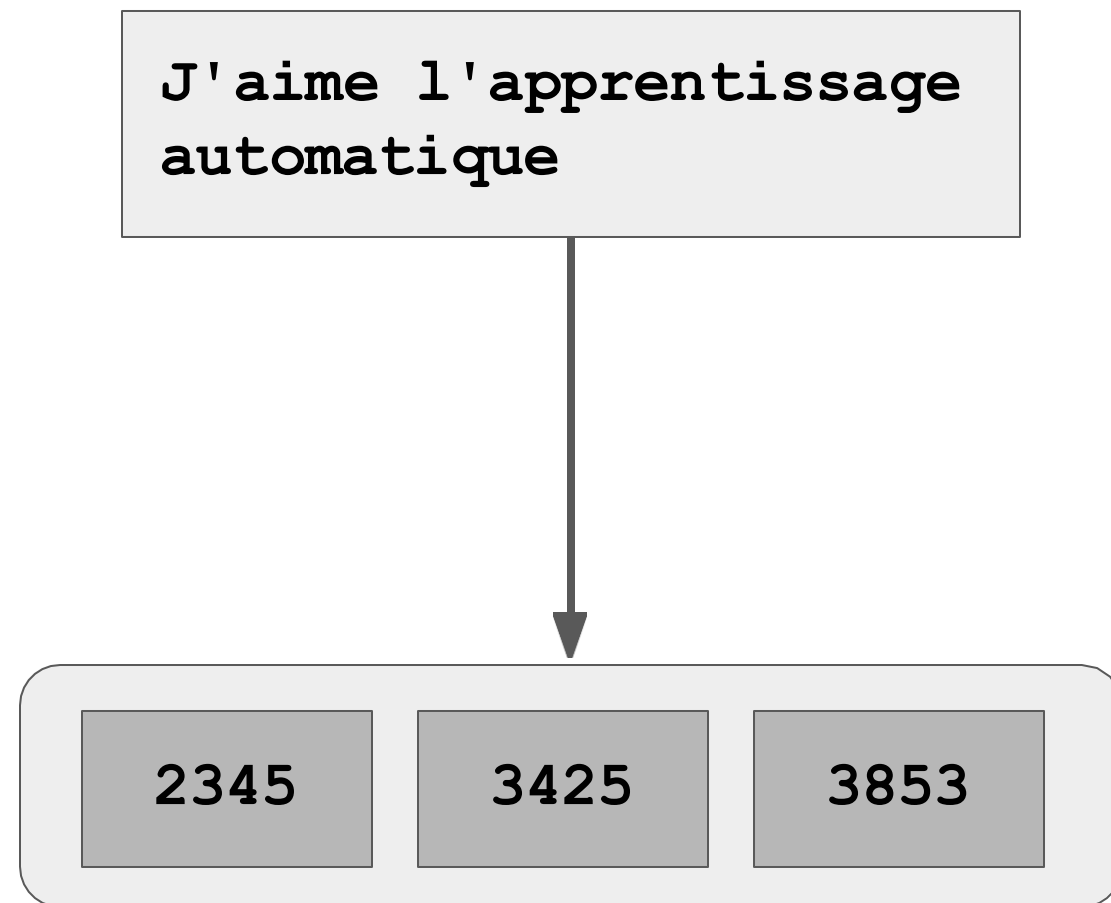
Soft Prompts

Prompt Tuning

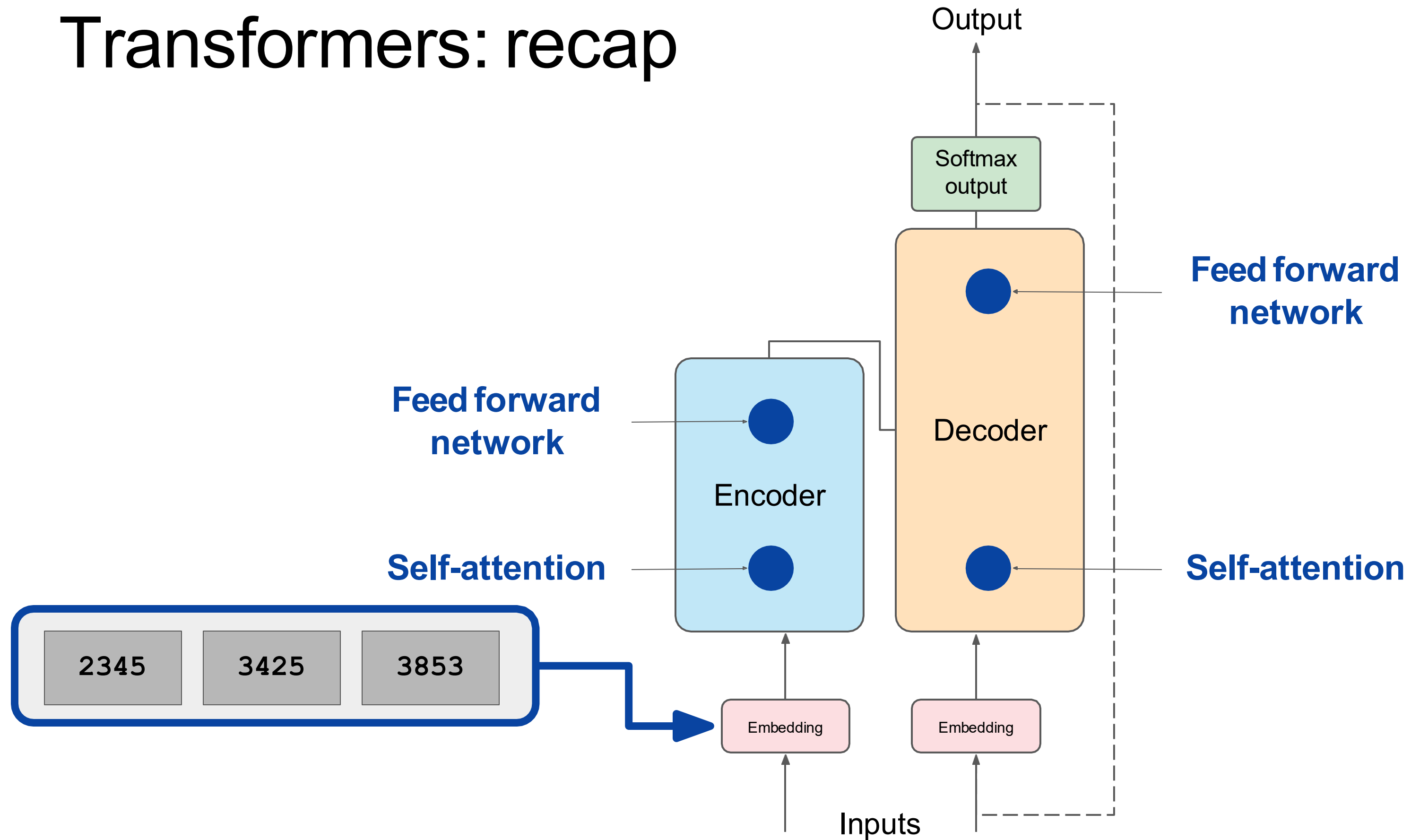
Source: Lialin et al. 2023, “Scaling Down to Scale Up: A Guide to Parameter-Efficient Fine-Tuning”,

Low-Rank Adaptation of Large Language Models (LoRA)

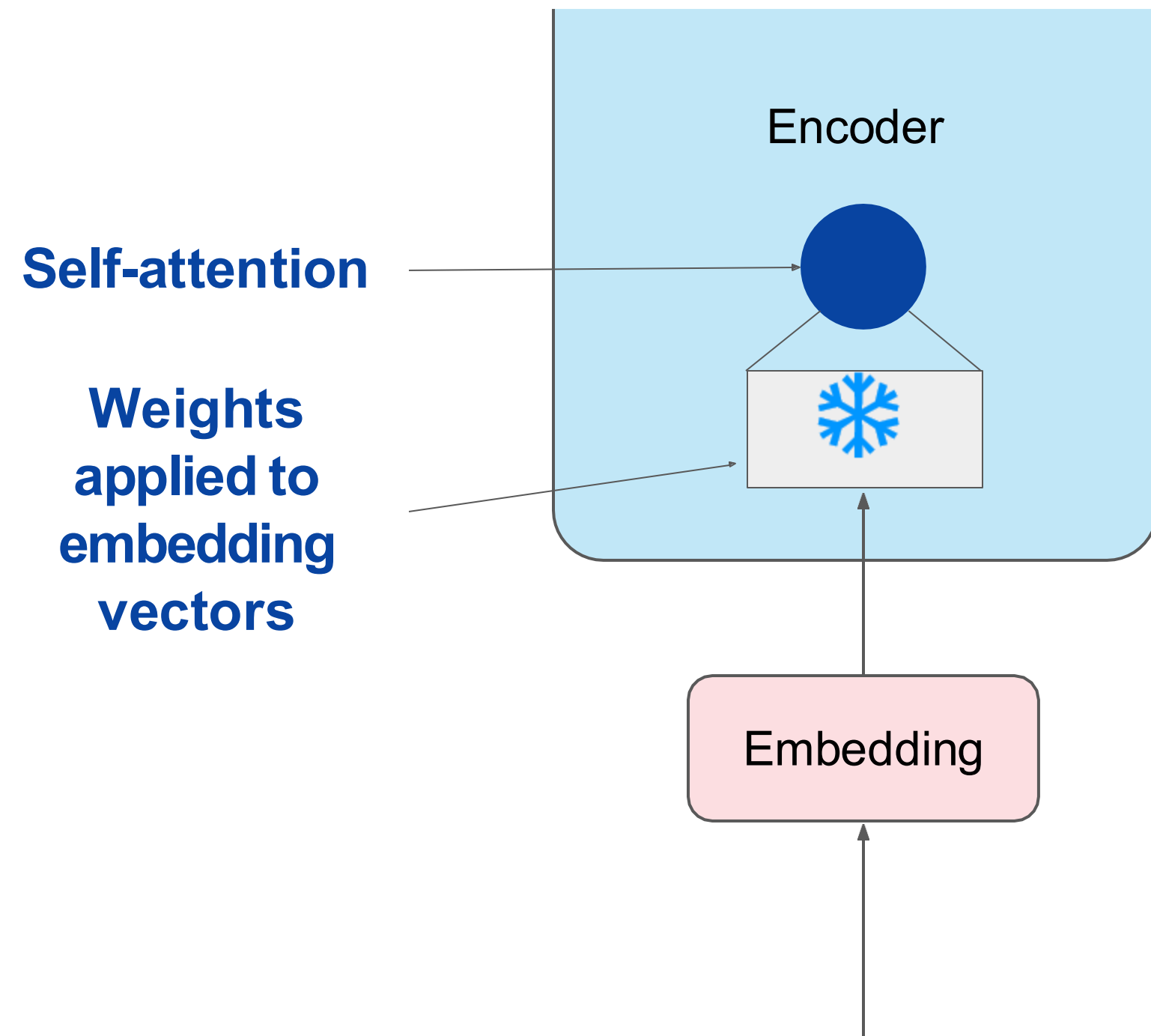
Transformers: recap



Transformers: recap

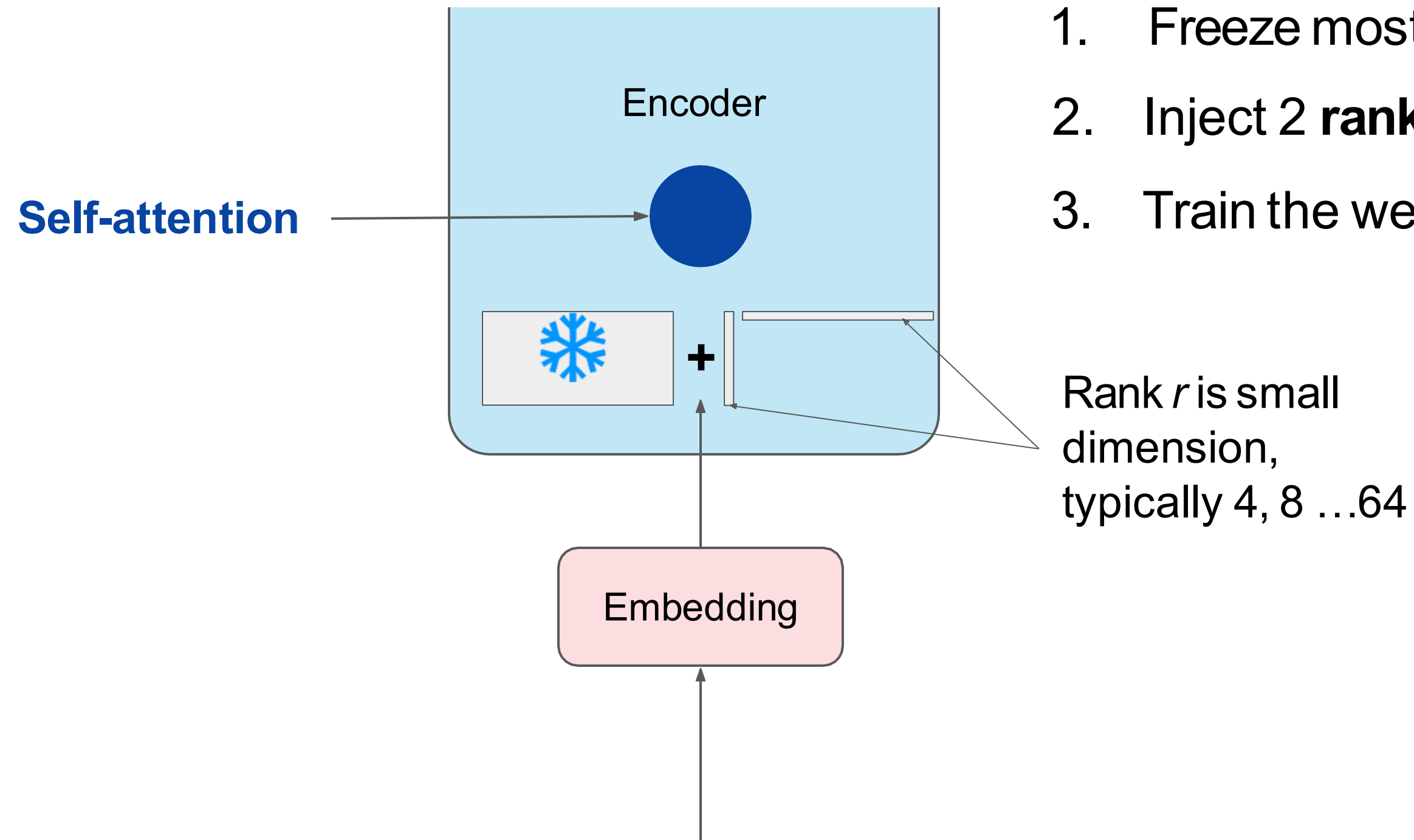


LoRA: Low Rank Adaption of LLMs



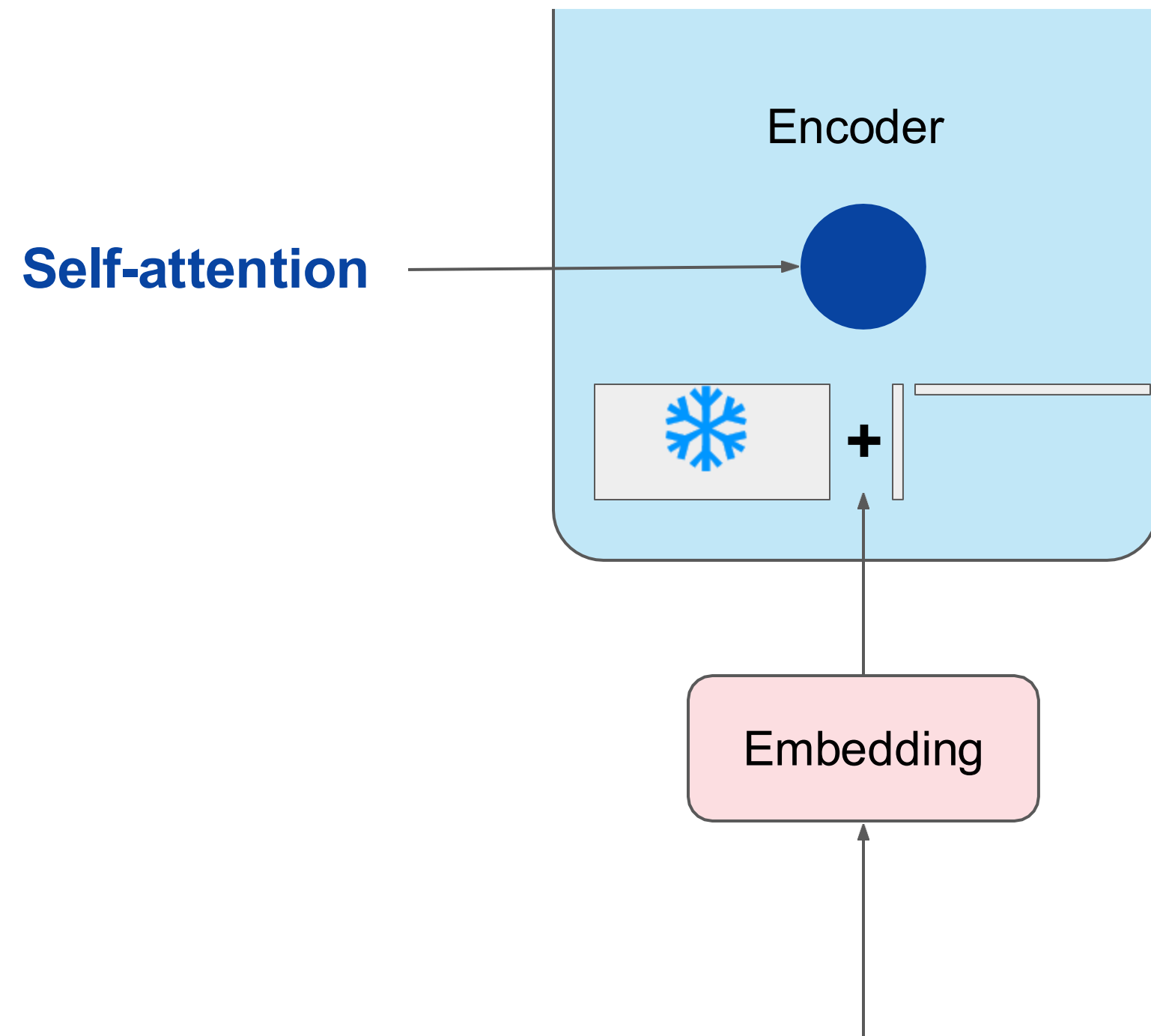
1. Freeze most of the original LLM weights.

LoRA: Low Rank Adaption of LLMs



1. Freeze most of the original LLM weights.
2. Inject 2 **rank decomposition matrices**
3. Train the weights of the smaller matrices

LoRA: Low Rank Adaption of LLMs



1. Freeze most of the original LLM weights.
2. Inject 2 **rank decomposition matrices**
3. Train the weights of the smaller matrices

Steps to update model for inference

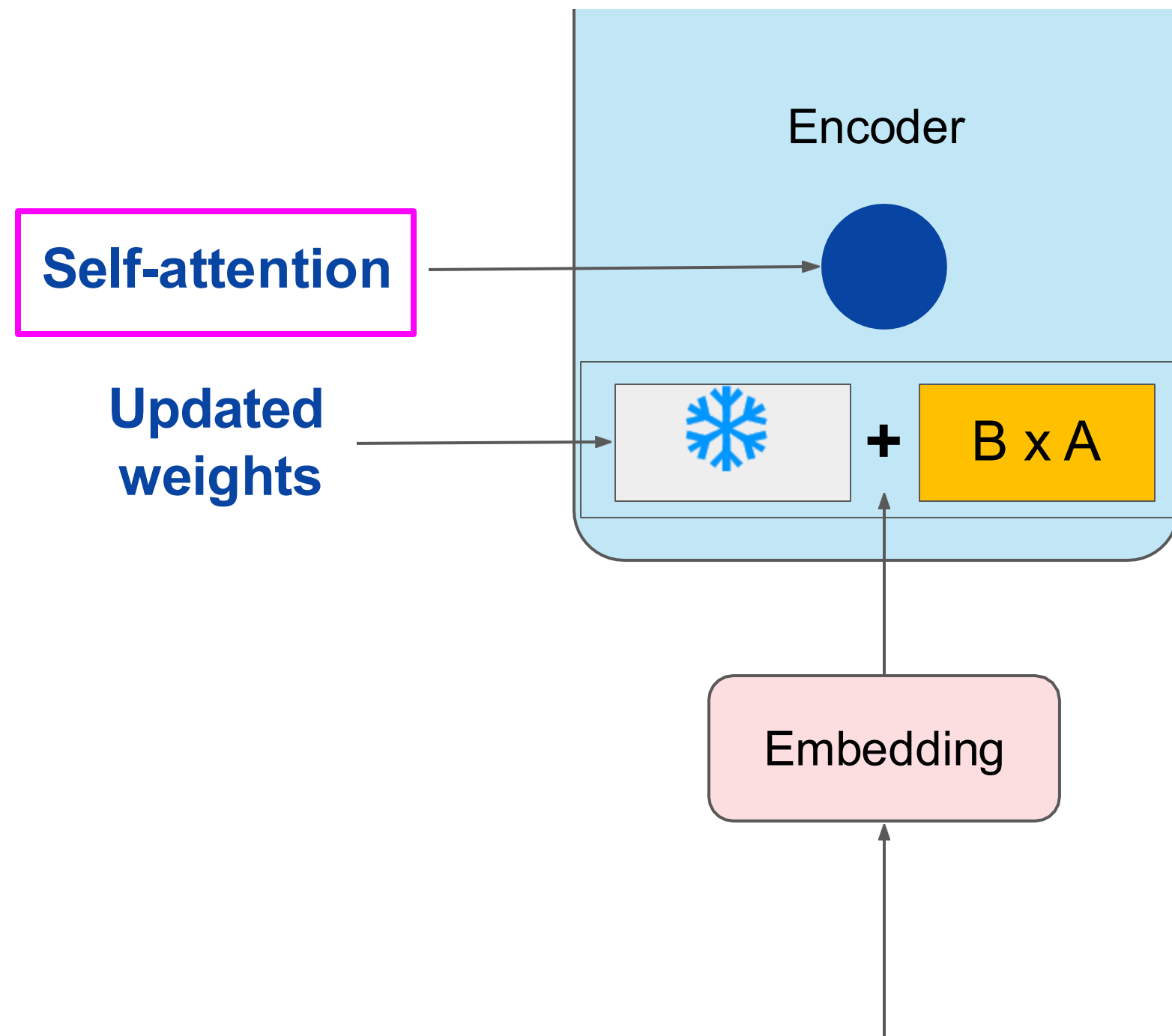
1. Matrix multiply the low rank matrices

$$\text{Matrix } B * \text{Matrix } A = \text{Matrix } B \times A$$

2. Add to original weights

$$\text{Matrix } B \times A + \text{Matrix } B \times A$$

LoRA: Low Rank Adaption of LLMs



1. Freeze most of the original LLM weights.
2. Inject 2 **rank decomposition matrices**
3. Train the weights of the smaller matrices

Steps to update model for inference:

1. Matrix multiply the low rank matrices

$$\begin{array}{|c|} B \end{array} * \overline{A} = \begin{array}{|c|} B \times A \end{array}$$

2. Add to original weights

$$\begin{array}{|c|} \text{Snowflake} \end{array} + \begin{array}{|c|} B \times A \end{array}$$

Concrete example using base Transformer as reference

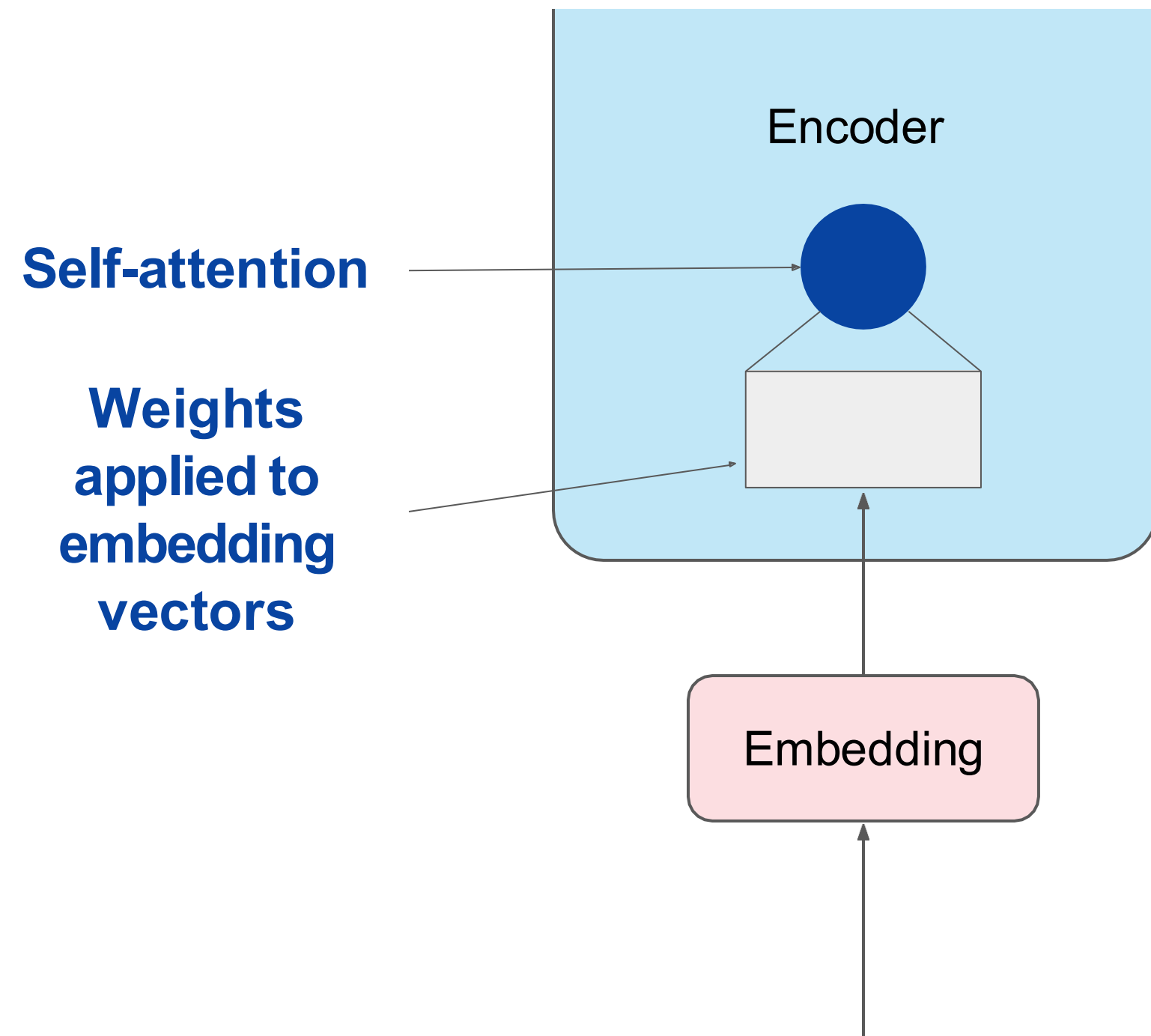
Use the base Transformer model presented by Vaswani et al. 2017:

- Transformer weights have dimensions $d \times k = 512 \times 64$
- So $512 \times 64 = 32,768$ trainable parameters

In LoRA with rank $r = 8$:

- A has dimensions $r \times k = 8 \times 64 = 512$ parameters
- B has dimension $d \times r = 512 \times 8 = 4,096$ trainable parameters
- **86% reduction in parameters to train!**

LoRA: Low Rank Adaption of LLMs

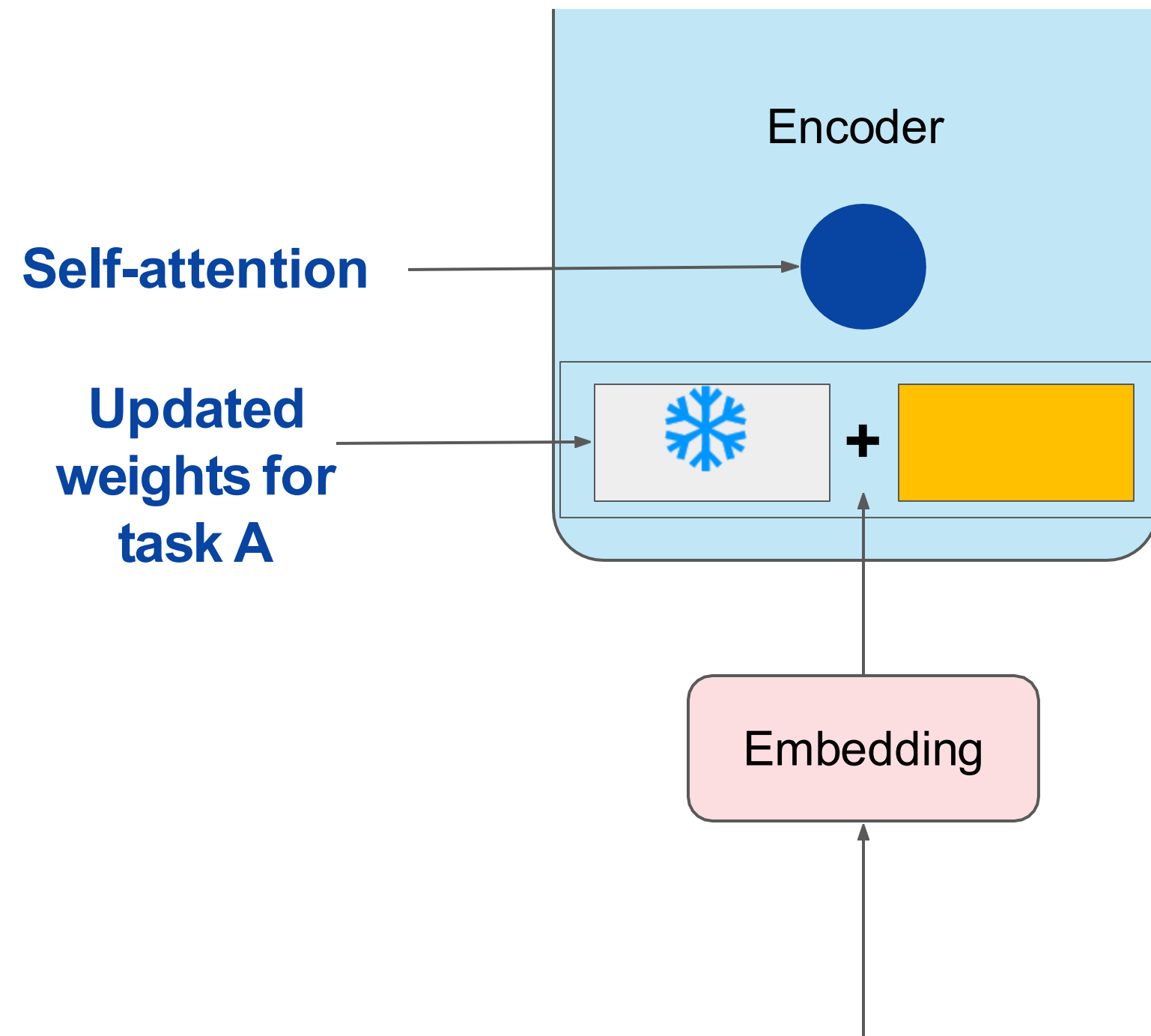


1. Train different rank decomposition matrices for different tasks
2. Update weights before inference

Task A

$$\begin{array}{c} \text{Yellow bar} \\ * \end{array} \begin{array}{c} \text{Yellow bar} \\ + \\ \begin{array}{c} \text{Gray box with snowflake} \\ + \\ \text{Yellow box} \end{array} \end{array} = \begin{array}{c} \text{Yellow box} \end{array}$$

LoRA: Low Rank Adaption of LLMs



1. Train different rank decomposition matrices for different tasks
2. Update weights before inference

Task A

$$\begin{array}{c} \text{Yellow bar} \\ * \end{array} \begin{array}{c} \text{Yellow bar} \\ * \end{array} = \begin{array}{c} \text{Yellow bar} \\ * \end{array}$$

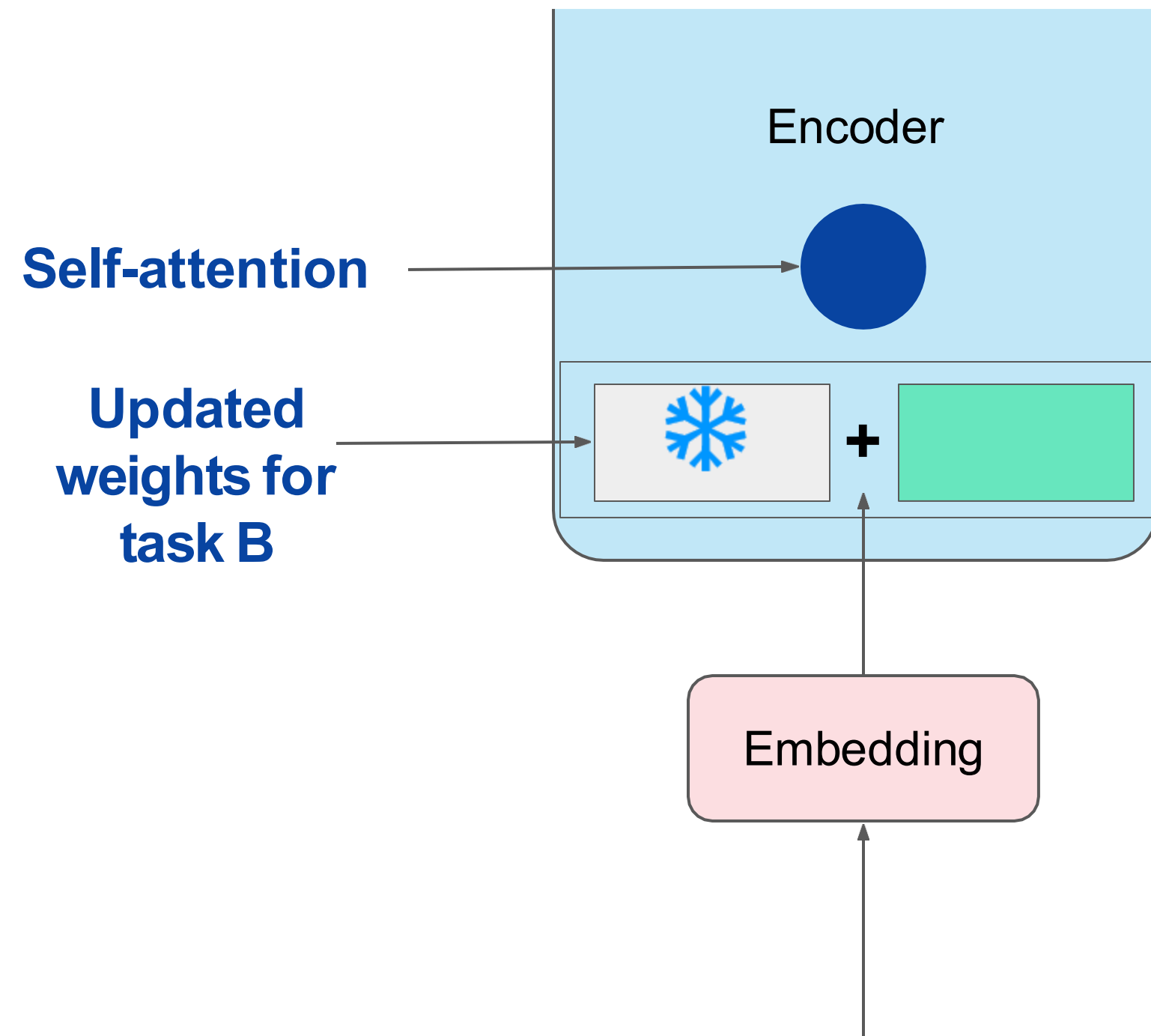
The diagram shows the weight update for Task A. It consists of a vertical yellow bar, a multiplication sign, a horizontal yellow bar, an equals sign, and a yellow rectangle. Below this, a box contains a light gray rectangle with a blue snowflake icon, a plus sign, and a yellow rectangle.

Task B

$$\begin{array}{c} \text{Teal bar} \\ * \end{array} \begin{array}{c} \text{Teal bar} \\ * \end{array} = \begin{array}{c} \text{Teal bar} \\ * \end{array}$$

The diagram shows the weight update for Task B. It consists of a vertical teal bar, a multiplication sign, a horizontal teal bar, an equals sign, and a teal rectangle. Below this, a box contains a light gray rectangle with a blue snowflake icon, a plus sign, and a teal rectangle.

LoRA: Low Rank Adaption of LLMs



1. Train different rank decomposition matrices for different tasks
2. Update weights before inference

Task A

$$\begin{array}{c} \text{Yellow bar} \\ * \\ \text{Yellow bar} \end{array} = \text{Yellow box}$$
$$\begin{array}{c} \text{Snowflake box} + \text{Yellow box} \end{array}$$

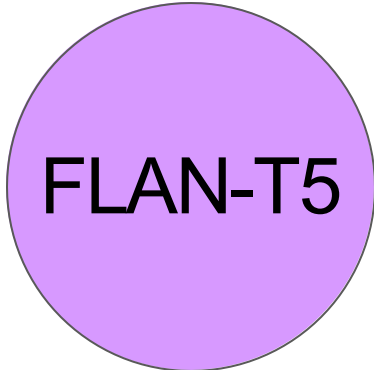
Task B

$$\begin{array}{c} \text{Green bar} \\ * \\ \text{Green bar} \end{array} = \text{Green box}$$
$$\begin{array}{c} \text{Snowflake box} + \text{Green box} \end{array}$$

Sample ROUGE metrics for full vs. LoRA fine-tuning

Base model
ROUGE

Full fine-tune
ROUGE

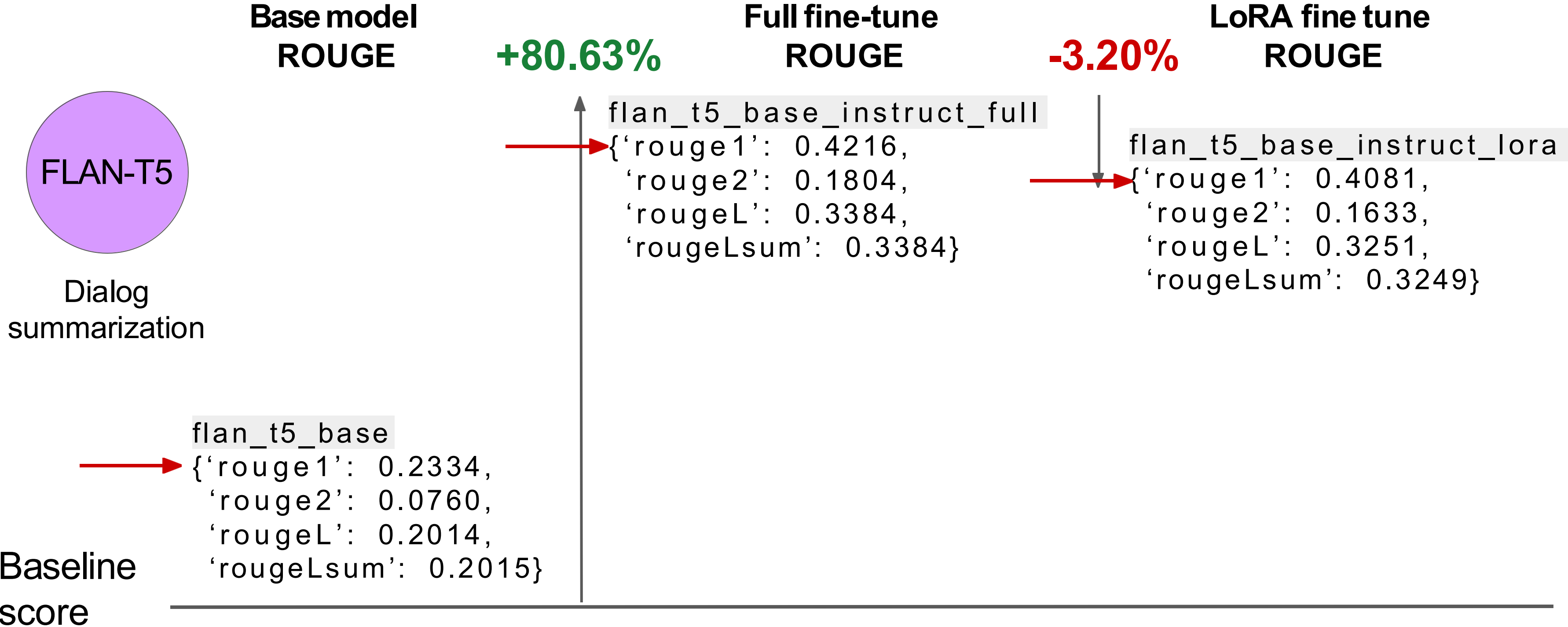


Dialog
summarization

→ `flan_t5_base`
{`'rouge1'`: 0.2334,
`'rouge2'`: 0.0760,
`'rougeL'`: 0.2014,
`'rougeLsum'`: 0.2015}

Baseline
score

Sample ROUGE metrics for full vs. LoRA fine-tuning



Choosing the LoRA rank

Rank r	val_loss	BLEU	NIST	METEOR	ROUGE_L	CIDEr
1	1.23	68.72	8.7215	0.4565	0.7052	2.4329
2	1.21	69.17	8.7413	0.4590	0.7052	2.4639
4	1.18	70.38	8.8439	0.4689	0.7186	2.5349
8	1.17	69.57	8.7457	0.4636	0.7196	2.5196
16	1.16	69.61	8.7483	0.4629	0.7177	2.4985
32	1.16	69.33	8.7736	0.4642	0.7105	2.5255
64	1.16	69.24	8.7174	0.4651	0.7180	2.5070
128	1.16	68.73	8.6718	0.4628	0.7127	2.5030
256	1.16	68.92	8.6982	0.4629	0.7128	2.5012
512	1.16	68.78	8.6857	0.4637	0.7128	2.5025
1024	1.17	69.37	8.7495	0.4659	0.7149	2.5090

- Effectiveness of higher rank appears to plateau
- Relationship between rank and dataset size needs more empirical data

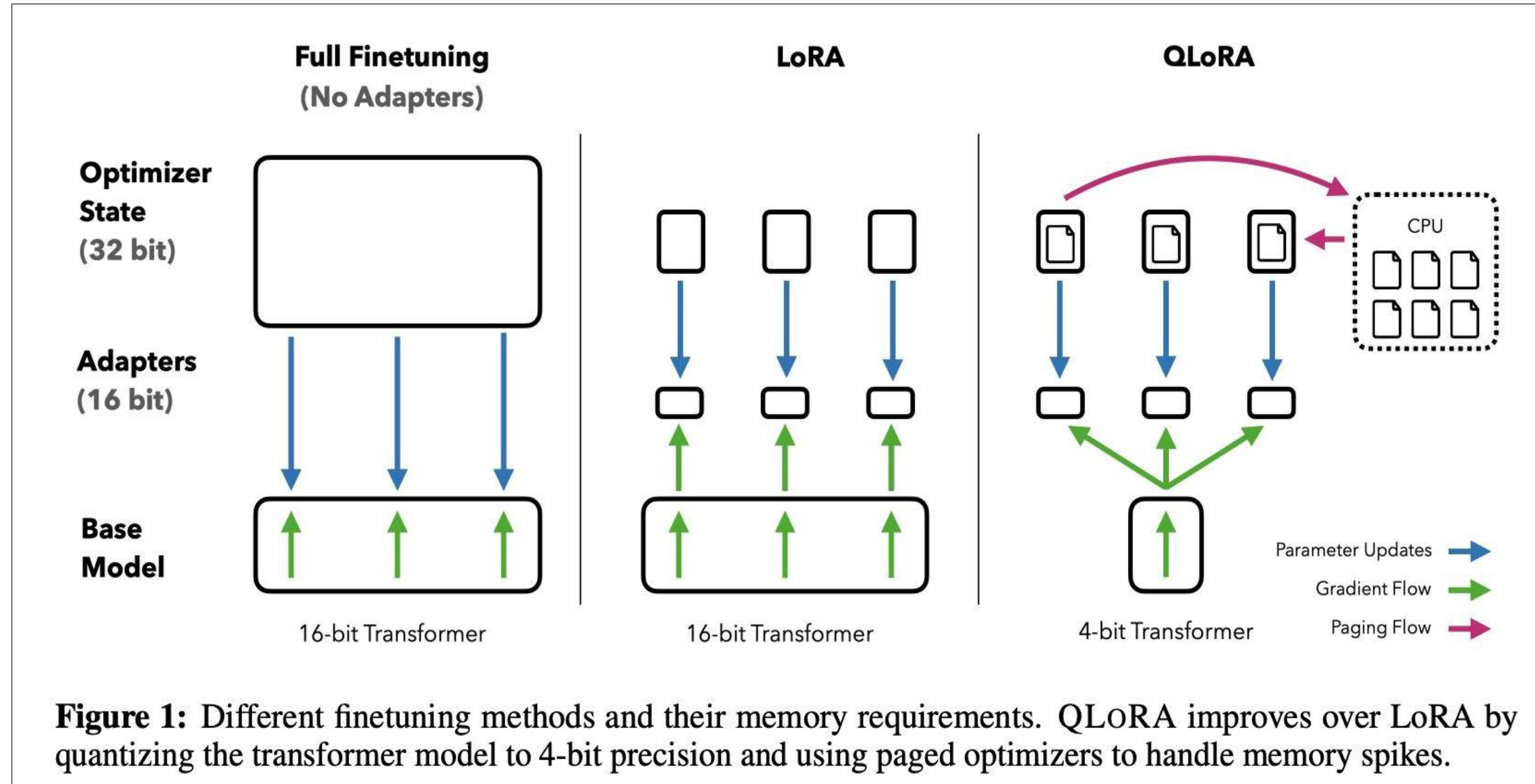
Source: Hu et al. 2021, “LoRA: Low-Rank Adaptation of Large Language Models”

QLoRA: Quantized LoRA

- Introduces 4-bit NormalFloat (nf4) data type for 4-bit quantization
- Supports double-quantization to reduce memory ~0.4 bits per parameter (~3 GB for a 65B model)
- Unified GPU-CPU memory management reduces GPU memory usage
- LoRA adapters at every layer - not just attention layers
- Minimizes accuracy trade-off

Source: Dettmers et al. 2023, "QLoRA: Efficient Finetuning of Quantized LLMs"

QLoRA: Quantized LoRA



Source: Dettmers et al. 2023, "QLoRA: Efficient Finetuning of Quantized LLMs"

QLoRA: Quantized LoRA

Method	Base Model	Trainable Parameters	Adapters Used	GPU Memory Usage	Key Advantage
Full Finetuning	16-bit	All parameters (entire model)	 None	 Very High	Maximum accuracy, full flexibility
LoRA (Low-Rank Adaptation)	16-bit	Only small adapter matrices	 Yes	 Moderate	Much lower memory use; faster training
QLoRA (Quantized LoRA)	4-bit (quantized)	Only adapters	 Yes	 Very Low	Enables fine-tuning huge models on one GPU; uses paged optimizer to handle memory spikes

PEFT methods summary

Selective

Select subset of initial LLM parameters to fine-tune

Reparameterization

Reparameterize model weights using a low-rank representation

LoRA

Additive

Add trainable layers or parameters to model

Adapters

Soft Prompts
Prompt Tuning

Lecture References

- Generative AI with Large Language Models – DeepLearning.ai Course (Week 2)